CoolProp CAPE-OPEN Property Package: Technical Reference

Release 0.3

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Implementation, design decisions, flash and property routing, transport-property fallbacks, and validation of the CoolProp-backed CAPE-OPEN 1.1 thermodynamic property package.

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Version	Date	Description
0.1	2026-07	Initial draft: architecture, design decisions, flash routing, transport, validation
0.2	2026-07	Air-separation unit (ASU) validation case and PFD
0.3	2026-10	CO_2 compression/liquefaction (CCS) validation case; re-solved on CoolProp 591e2eff (package 1.1.5)

Part I

Foundations

Chapter 1

Introduction

1.1 Purpose and scope

This reference documents the **CoolProp CAPE-OPEN property package**: a CAPE-OPEN 1.1 [1] thermodynamic property package that wraps the open-source **CoolProp** library [2] and exposes it to process-modelling environments (PMEs) such as COCO/COFE and ChemSep. The package makes CoolProp's Helmholtz-explicit multiparameter equations of state (the `HEOS` backend — a multi-fluid Helmholtz-energy mixture model in the tradition of [3] and [4], with departure functions in the GERG-2008 style of [5]) and its cubic backends (`PR`, `SRK`) available as a selectable, multi-compound property package with a full vapour–liquid equilibrium flash, single-phase caloric/volumetric/transport properties, and persistence in the flowsheet file.

The package is an **interface layer**, not a thermodynamic engine. Following the CAPE-OPEN adapter pattern, it translates PME requests into calls to CoolProp's low-level `AbstractState` C interface and maps the results back onto CAPE-OPEN property identifiers and SI units (*Architecture*). All of the equation-of-state mathematics — reducing functions, departure functions, density solving, the underlying multiparameter correlations — belongs to CoolProp and is documented in the CoolProp literature [2]; **this reference deliberately does not reproduce it.** What it *does* document is everything that lives in the CAPE-OPEN layer and its thin calculation shim: the design decisions, the flash and property routing, the fallbacks that make the package robust inside a real flowsheet solve, and the validation of that behaviour on three reference process flowsheets.

> **Note**
>
> Throughout, "the package" means the CAPE-OPEN wrapper and its C# calculation shim (`CoolPropCO.CapeOpen` + `CoolPropCO.Core`); "CoolProp" means the upstream native library reached through the `AbstractState` handle.

1.2 What made this possible: the upstream mixture flash

A CAPE-OPEN package is only as usable as its flash is robust. A PME evaluates a property package *thousands* of times per converged flowsheet — at feasible states, at wildly infeasible recycle guesses, at the exact saturation boundary, and on degenerate (single-component, zero-fraction) compositions — and a single unrecoverable failure aborts the whole solve.

For most of CoolProp's history its **mixture** flash was not robust enough for this. The historical `HEOS` mixture `update(PT_INPUTS, ...)` could return a metastable single-phase density root at or near a

phase boundary, fail its successive-substitution / density solve a hair from convergence, or silently converge a state-function (PH/PS) flash to the wrong temperature. Any of these turns a routine flowsheet evaluation into a non-physical result or a hard stop.

Recent upstream improvements to the CoolProp mixture flash are what make this package practical. The reworked `HSU_P` state-function flash, the more reliable phase-split and stability handling, and the improved saturation (`PQ`/`QT`) solvers upstream mean that, for the first time, CoolProp's mixture routines converge reliably enough across the state space that a CAPE-OPEN package can lean on them — with a thin, verified safety net rather than a full re-implementation of the flash. The validation flowsheets in *Validation* — a steam cycle, and two mixed-refrigerant LNG trains solved end-to-end in COFE against the cubic reference — are a direct demonstration of that maturity.

1.3 What this reference covers

- the **software architecture** and the `AbstractState`-handle model, including the dedicated equilibrium / liquid / vapour handles (*Architecture*);
- the **design decisions** that keep the package robust and correct in a PME: compound-order alignment by CAS, zero-fraction handling, single-component (pure-fluid) routing, and EOS/backend selection (*Design Decisions*);
- the **flash and property routing**: TP, the robust PH/PS strategy and the optional native `HSU_P` path, saturation flashes, and the micro-perturbation convergence retry (*Flash and Property Routing*);
- the **transport-property fallbacks** used when CoolProp has no viscosity/conductivity/surface-tension model for the selected fluid or backend — as is common on the cubic backends (*Transport-Property Fallbacks*);
- the **incompressible-fluid** companion package for utility streams (*The Incompressible Companion Package*);
- **validation** of the package on the CHP, TEALARC, and Snøhvit flowsheets in COFE against the cubic reference (*Validation*);
- an **options and identifiers** reference for configuration and property support (*Appendix: Options and Property Identifiers*).

1.4 Audience

The reader is assumed to be familiar with process thermodynamics, the vapour –liquid equilibrium flash, and the CAPE-OPEN 1.1 thermodynamic interfaces. The CoolProp equations of state themselves are treated as a trusted dependency.

Chapter 2

Architecture

2.1 The adapter pattern

The package follows the standard CAPE-OPEN adapter pattern [1]: the CAPE-OPEN COM layer contains **no thermodynamic calculation logic**. It reads T/P/composition from the PME's Material Object, calls into CoolProp for every actual EOS/flash/transport calculation, and writes the results back onto the Material Object in CAPE-OPEN's units and property identifiers.

```
PME (COFE / ChemSep)
  |
  v
Material Object                          (implemented by the PME)
  |  ICapeThermoMaterial, ICapeThermoPhases, ICapeThermoCompounds
  v
CoolPropCO.CapeOpen                      (the interface layer - this package)
  |  ICapeThermoPropertyRoutine, ICapeThermoEquilibriumRoutine,
  |  ICapeThermoCompounds, ICapeThermoPhases, IPersistStreamInit, ...
  v
CoolPropCO.Core                          (thin calculation shim)
  |  ICoolPropEngine: Configure / Flash / CalcSinglePhaseProps /␣
↪SurfaceTension
  v
CoolProp (native)                        (AbstractState C interface, P/Invoke)
```

Three projects implement this:

Project	Responsibility
`CoolPropCO.CapeOpen`	The COM-visible property package (net48, x64). Implements every CAPE-OPEN 1.1 interface, the `MaterialObjectWrapper` (safe VARIANT coercion), persistence, logging, and COM registration.
`CoolPropCO.Core`	Compound catalog, models (`Component`, `FlashSpec`, `FlashResult`, `EosModel`, `PhPsStrategy`), the `ICoolPropEngine` seam, and the real P/Invoke engine (`CoolPropEngine`) plus a non-calculating `StubEngine` used before the engine is wired or for interface-only testing.
`CoolPropCO.Editor`	The Avalonia 11 configuration GUI shown by `Edit()` (compound selection, EOS, PH/PS strategy, kij overrides), and the companion incompressible-fluid editor.

`ICoolPropEngine` is the single seam between the CAPE-OPEN layer and CoolProp: the CAPE-OPEN code depends only on this interface, never on CoolProp's native API directly. Native interop is flat P/Invoke against CoolProp's integer-handle `AbstractState_*` C interface — no C++/CLI bridge and no SWIG wrapper, which avoids the mixed-mode COM-activation issues those would introduce inside a PME's process.

2.2 The AbstractState handle model

CoolProp's low-level interface centers on an **AbstractState**: an opaque integer handle representing one EOS/backend/fluid-set combination, updated in place by successive `update(input_pair, v1, v2)` calls. Each update caches the converged state (density, phase) as the density guess for the *next* update on that same handle — this is what makes repeated updates on a warm handle fast, but it also means **a handle's history can bias a later result** if the handle is reused for calls with different intent.

The engine keeps **four kinds of handle**, each with a narrow, fixed purpose, precisely to keep that history from crossing between intents:

Handle	Purpose
`_handle`	The **equilibrium** handle. Used for every `CalcEquilibrium` flash (TP, PH, PS, PVF, TVF) and nothing else. Never phase-imposed.
`_liqHandle`	Single-phase property calls (`CalcSinglePhaseProp`) for a phase the PME has told us is liquid. Phase-imposed (`specify_phase("phase_liquid")`) for the duration of the call, then unimposed.
`_vapHandle`	The vapour analogue of `_liqHandle`.
`_pure[i]`	One pure-fluid `AbstractState` per configured component, used when the composition handed to the engine is (numerically) pure in component *i*. See *Design Decisions* for why this exists.

Why separate liquid/vapour handles from the equilibrium handle. A single-phase property call *imposes* a phase on its handle (`specify_phase`) so CoolProp's density solver doesn't have to rediscover which root is wanted. If that same handle were later reused for an *unimposed* equilibrium flash, the leftover imposed-phase state and density guess can bias `update(PT_INPUTS, ...)` onto the wrong (sometimes **metastable**) root for a state that is genuinely two-phase. This was observed directly during validation on the TEALARC flowsheet's C2/C3 condenser: a state that is unambiguously two-phase ($Q \approx 0.4$) in isolation came back as all-liquid inside the live flowsheet solve. Giving the equilibrium flash its own handle — one that is *never* phase-imposed and never touched by a single-phase property call — removes that specific cross- contamination path. (For the record: in that particular investigation the actual root cause turned out to be a compound-order mismatch, not handle contamination — see *Design Decisions* — but the handle separation is still the correct defensive design and is kept regardless, since it eliminates a real class of cross-pollution between the imposed-phase and equilibrium code paths.)

Reconfiguration. `Configure(eos, components, kijOverrides, phPsStrategy)` frees any existing handles and rebuilds all of them — the four equilibrium/ liquid/vapour/pure handles — from the current compound list, applying any kij (binary interaction parameter) overrides keyed by CAS pair. It is called whenever the EOS, compound selection, or compound *order* changes (the last of these matters — see *Design Decisions*), and idempotently on every `Edit()` confirmation and flowsheet `Load()`.

2.3 Pure-fluid routing

CoolProp's **mixture** backend does not automatically route a degenerate (single-component) composition to a pure-fluid solver — it still runs the full mixture flash machinery, which can fail outright on some pure compositions (observed: a mixture handle throws on certain pure water states that a dedicated pure handle solves without issue, and pure-handle evaluation is measured to be roughly 40x faster). Since a PME routinely evaluates pure compositions — CAPE-OPEN's reference-state harmonisation pass (§5.6) evaluates every compound individually — the engine detects an exactly pure composition (one fraction $\geq 1 - 10^{-9}$, all others $\leq 10^{-9}$) and routes it to the matching `_pure[i]` handle instead of the mixture handle. Near-pure trace compositions (a small but nonzero secondary component) deliberately stay on the mixture path, so trace tracking through a flowsheet is not silently short-circuited.

2.4 COM surface and lifetime

The package implements the full CAPE-OPEN 1.1 interface set required of a property package (*Introduction*): `ICapeThermoPropertyRoutine`, `ICapeThermoEquilibriumRoutine`, `ICapeThermoMaterialContext`, `ICapeThermoCompounds`, `ICapeThermoPhases`, `ICapeThermoUniversalConstants`, `ICapeIdentification`, `ICapeUtilities`, and `IPersistStreamInit` for flowsheet persistence. COM lifetime follows the pattern the CAPE-OPEN skill's field notes document as correct: the only external COM references the package holds — the Material Object and the simulation context — are released via `Marshal.ReleaseComObject` in `UnsetMaterial/SetMaterial` (on replacement) and `Terminate`, with a residual-reference-count sentinel logged if a release is ever incomplete. The `IStream` handed to `Save/Load` is the PME's, not ours, so it is wrapped with a no-op `Dispose`.

Two GUIDs identify the package: the Property Package CLSID (`E5A65162-CFB9-4394-8AAC-FAEA7239E420`, ProgId `CoolPropCO.PropertyPackage`) and the Package Manager CLSID (`FE29F896-33CA-4F0D-9147-2AFB91B7550D`, ProgId `CoolPropCO.PackageManager`); the companion incompressible package (*The Incompressible Companion Package*) has its own CLSID (`837C291D-A3C5-4E4E-A981-448A6F0896FA`, ProgId `CoolPropCO.IncompressiblePackage`). Registration is per-user (HKCU), no administrator rights required.

2.5 Configuration GUI

`Edit()` runs the Avalonia configuration dialog on a dedicated, long-lived STA thread separate from the PME's own COM STA, and blocks the calling thread until the dialog closes. Two robustness measures matter for a COM-embedded GUI specifically:

- **Software rendering is forced.** GPU/ANGLE/OpenGL initialisation can hang indefinitely on remote-desktop sessions, VMs, and machines without GPU drivers — none of which a configuration dialog needs.
- **The UI-thread startup wait is bounded** (30 s). If the UI stack fails or stalls, `Edit()` logs the failure and returns gracefully instead of blocking the PME forever.

2.6 Registered as a selectable property package

Once COM-registered, the package appears in the PME's own package chooser alongside every other CAPE-OPEN 1.1 package on the system — evidence that COM registration and the CAPE-OPEN category GUIDs (*Architecture*, above) are correctly wired end-to-end, not just that the package builds.

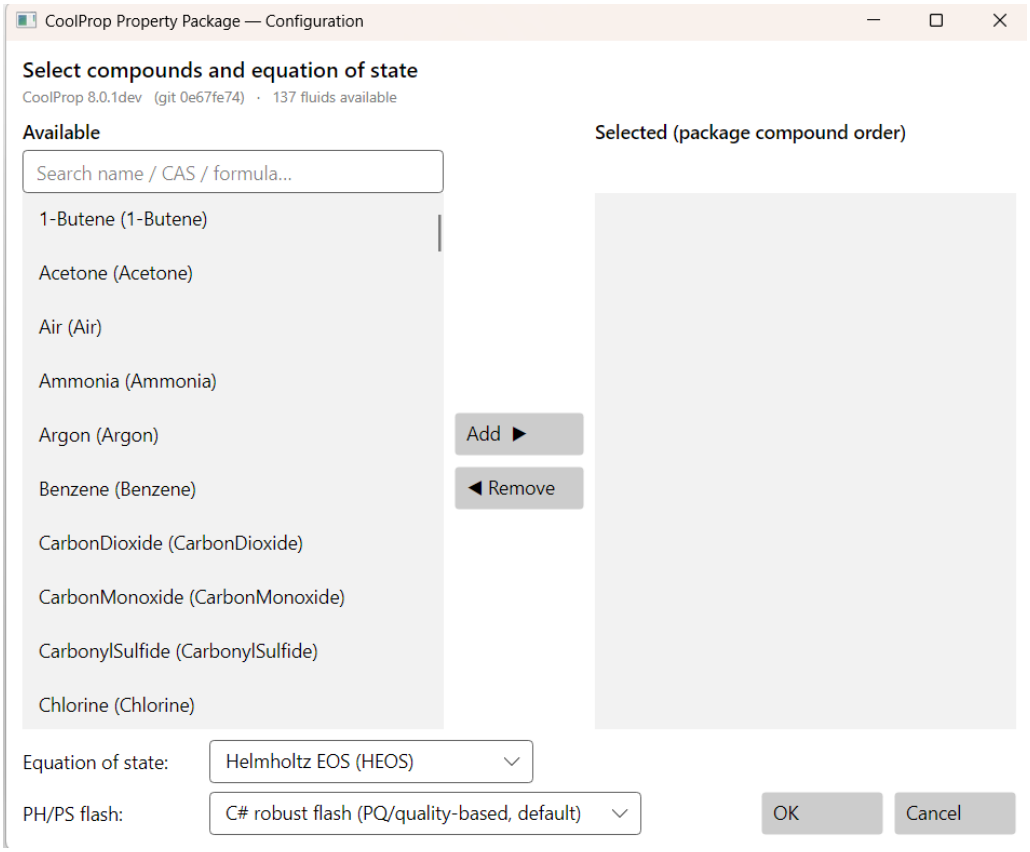


Figure 2.1: The compound-selection / EOS / PH/PS-strategy configuration dialog shown by `Edit()`. See *Design Decisions* for the PH/PS strategy control this dialog exposes.

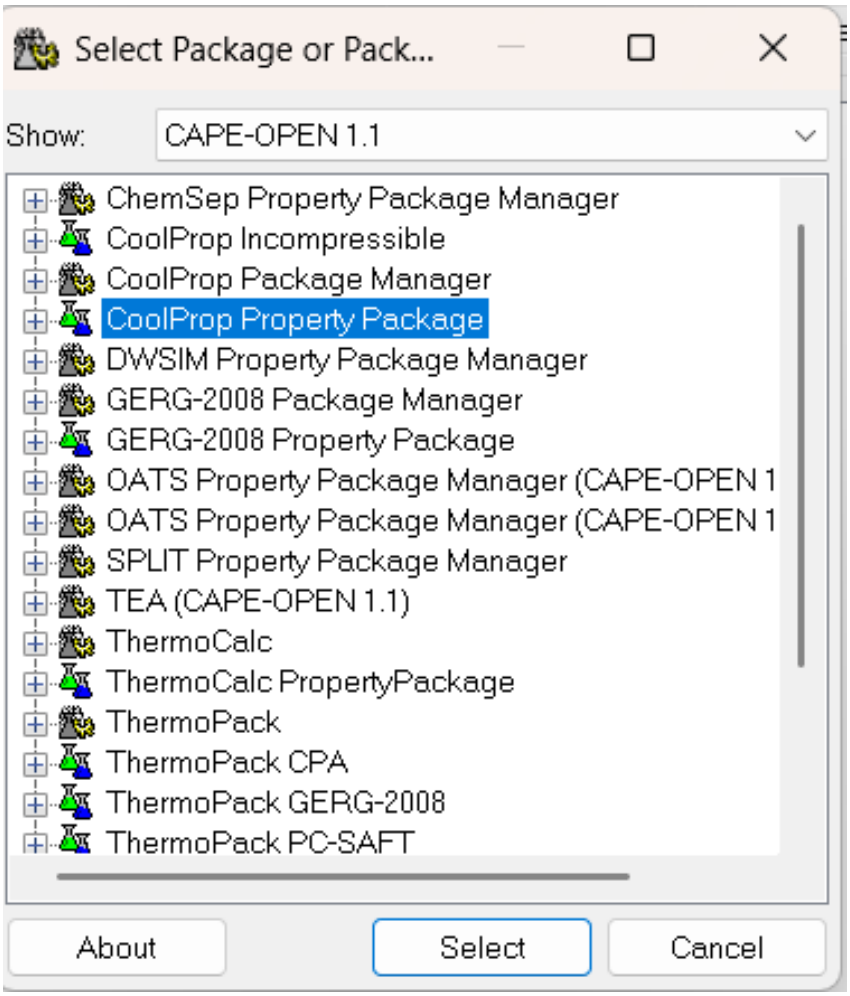


Figure 2.2: COFE's "Select Package or Package Manager" dialog, showing the CoolProp Property Package (and the companion CoolProp Incompressible package) registered alongside GERG-2008, ThermoCalc, ChemSep, TEA, and the other CAPE-OPEN 1.1 packages installed on the system.

2.7 Persistence

Configuration is saved into the flowsheet file via `IPersistStreamInit` as a small versioned binary stream: format version, EOS selection, the ordered compound list (CoolProp name + CAS per compound), kij overrides (keyed `"CAS1|CAS2"`), and the PH/PS strategy. The stream version lets the format grow — a v1 stream (saved before the PH/PS strategy field existed) loads cleanly with that field defaulting to the C# strategy. Because COFE creates a fresh property-package COM instance per solve cycle, the *last-edited* configuration is also broadcast through static fields shared across instances within one PME session, so instances created mid-solve (without their own `Load()`) pick up the same compounds/EOS/strategy as the instance the user configured via `Edit()`.

Part II

Implementation and Design Decisions

Chapter 3

Design Decisions

This chapter documents the decisions in the interface layer that are not obvious from the CAPE-OPEN spec alone: how zero/near-zero compositions are handled, how single-component ("pure") states are routed, how compound order is kept consistent with the PME, how the EOS/backend is selected, and how the PH/PS flash strategy is chosen. Each of these was driven by a concrete flowsheet failure mode, not a hypothetical one — the failure is described alongside the fix.

3.1 Zero-fraction components

A property package is normally configured with more compounds than any one stream actually uses — a 5-compound refrigerant package serving a 2-compound refrigerant loop, for example. CAPE-OPEN composition arrays are dense (one entry per configured compound, in a fixed order — see *Compound-order alignment to the Material Object*), so the unused compounds arrive as **exact zeros**, not as absent entries.

CoolProp's `HEOS` mixture reducing and departure functions divide by composition-weighted terms. An exact-zero component makes those terms singular, and the density solver returns a non-physical result (`solver_rho_Tp ... p is not a valid number`) — for *every* flash on that stream, at *any* T/P, not just at edge cases. This was root-caused on the TEALARC flowsheet's methane/ethane refrigerant loop, configured against a 5-compound package: a feed of `[0.6, 0.4, 0, 0, 0]` fails outright, while flooring the three zeros to 10^{-12}–10^{-8} and renormalizing succeeds and reproduces a true 2-compound package's result to 10 significant figures.

The fix, applied uniformly wherever a composition is handed to CoolProp (`Flash`, `CalcSinglePhaseProps`, `SurfaceTension`): any mole fraction below a floor of 10^{-10} is lifted to 10^{-10} and the vector is renormalized before `set_fractions`.

$$x_i \leftarrow \max(x_i, 10^{-10}), \qquad x_i \leftarrow x_i \Big/ \sum_j x_j \tag{3.1}$$

10^{-10} is small enough to be thermodynamically negligible (many orders of magnitude below any flash convergence tolerance) while keeping the mixture's reducing functions well-conditioned. **Implementation**: `CoolPropEngine.SetFractionsSafe()`, `CompFloor = 1e-10`.

> **Note**
>
> This is distinct from the *exactly*-pure case below, which is not an edge case of the same clamp — an exactly pure composition is routed to a different (pure-fluid) handle entirely, not clamped and passed to the mixture backend.

3.2 Single-component (pure) compositions

The full mechanism and rationale are described in *The AbstractState handle model* (*Architecture*) alongside the rest of the handle model; summarized here as a design decision: CoolProp's **mixture** `AbstractState` backend does not itself detect and short-circuit to a pure-fluid solver when handed a degenerate one-component composition, and can fail outright on pure compositions the dedicated pure-fluid solver handles without issue (observed on pure water at the reference-state conditions COFE evaluates on package import). Since a PME evaluates pure compositions routinely — not only when a stream happens to be single-component, but as part of CAPE-OPEN's mandatory reference-state pass over every configured compound — the engine maintains one pure `AbstractState` per configured compound (`_pure[i]`) built at `Configure()` time, and routes an **exactly** pure composition ($x_i \geq 1 - 10^{-9}$, all others $\leq 10^{-9}$) to that handle instead of the mixture handle.

The exact-purity gate is deliberate: a composition that is *nearly* pure but carries a genuine trace of a second component stays on the mixture path, so trace-component tracking through a flowsheet (e.g. a ppm-level contaminant) is never silently dropped by snapping to the pure solver.

Pure-handle routing is also materially faster — CoolProp's dedicated pure-fluid saturation/density solvers measured roughly 40x faster than the mixture backend evaluated at the same (degenerate) composition — which matters because the reference-state pass touches every compound on every package instantiation.

3.3 Compound-order alignment to the Material Object

CAPE-OPEN composition arrays are **positional**: element i is "the i-th compound," with no identity carried alongside it. The order is established once, by whichever side of the interface the PME trusts as ground truth for a given material — and that order is **not guaranteed to match the property package's own saved compound order**. A flowsheet's compound list can be re-ordered by the PME's own bookkeeping, restored from a saved file whose order predates a later `Edit()`, or simply differ from whatever order the package happened to build its selection in.

If the two orders disagree, the feed **silently scrambles**: the mole fraction the PME intends for compound A lands on the package's compound B, and the flash returns a plausible-looking but wrong result — not an error. This was root-caused as the underlying cause of an extended investigation into why the TEALARC and Snøhvit flowsheets' CoolProp-backed results differed materially from the native cubic reference (*TEALARC: dual mixed refrigerant, the C2/C3 case study*, *Snøhvit: single mixed refrigerant, multi-stage compression*): the package had persisted its compound selection as `[Methane, Ethane, n-Butane, n-Propane, Nitrogen]` while the flowsheet's own order was `[Methane, Ethane, Propane, N-butane, Nitrogen]` — propane and n-butane transposed. Propane's mole fraction was therefore flashed as n-butane throughout the flowsheet; at the C2/C3 refrigerant condenser (50/50, 30 °C, 22.5 bar) that turns a genuinely two-phase state ($Q \approx 0.40$ for ethane/propane) into an all-liquid one (ethane/n-butane *is* liquid there), which cascaded into heat-exchanger temperature-limit warnings throughout the downstream cold end.

The fix follows directly from the CAPE-OPEN 1.1 specification (§7 — `ICapeThermoCompounds`) and is unambiguous once stated: the Material Object is the PME's own authority on a stream's compound order, so the package should adopt it rather than assume agreement. Before every positional composition read — in both `CalcEquilibrium` and `CalcSinglePhaseProp` — the package now:

1. reads the material's own compound list via `material as ICapeThermoCompounds -> GetCompoundList()` (`MaterialObjectWrapper.GetCompoundOrder()`);
2. resolves each entry to the package's catalog **by CAS number** (the stable identity key — see below);
3. if every compound resolves and the order differs from the package's current selection, adopts the

material's order as authoritative (`AlignSelectionToMaterial`), and the engine is reconfigured (handles rebuilt) the next time it is used.

This makes the package's own internal/persisted order irrelevant to correctness — it is realigned to the PME's ground truth before it can ever matter — and it is a no-op (zero reconfiguration cost) whenever the two orders already agree, which is the common case. It also degrades gracefully: if the material does not expose `ICapeThermoCompounds`, or any compound fails to resolve, the package falls back to its existing positional order rather than risk an incorrect realignment.

Why CAS, not name. CAS registry number is the one identifier CAPE-OPEN guarantees is stable and comparable across packages and PMEs; display names and formulae are not (a package's catalog and the PME's compound database can spell the same compound differently — "n-Butane" vs. "N-butane" is exactly the pair that triggered this investigation). The package's compound catalog carries a verified CAS number for every fluid it exposes (sourced from CoolProp's own `get_fluid_param_string(fluid, "CAS")`), so CAS-keyed resolution is reliable in both directions.

3.4 Equation-of-state / backend selection

One property package instance, one EOS at a time, selectable in the `Edit()` GUI and persisted with the flowsheet (not separate packages per EOS — a flowsheet swaps the EOS by editing the one package, not by rewiring which package a stream uses).

Selection	CoolProp backend	Notes
Peng-Robinson	`PR`	Cubic. No native transport model — see *Transport-Property Fallbacks* for the fallback correlations this triggers.
Soave-Redlich-Kwong	`SRK`	Cubic. Same transport-fallback behaviour as PR.
Helmholtz EOS	`HEOS`	CoolProp's multi-fluid Helmholtz-energy mixture model (*Introduction*). Highest accuracy; best native transport coverage; the default.

The three backends are validated against each other and against the native cubic reference in *Validation*.

3.4.1 Cold-start initialization on hard mixtures

HEOS's mixture density solver is, in general, more accurate but less uniformly robust across a wide composition/temperature range than a cubic EOS — the cubics were built for exactly this kind of numerical robustness across a broad flash-initialization search, at some cost in accuracy near difficult regions (e.g. an azeotrope). This shows up concretely in a PME's **initial-guess search**: when a flowsheet is solved from a fully cold start (no saved converged state to warm-start from — e.g. after a manual Reset), the PME's search can probe extreme trial compositions a converged solve never visits, and HEOS's mixture solver can fail outright on one of those trial points even when the flowsheet's actual converged solution is well within HEOS's normal operating range. This was observed directly on the CO2-from-NG extractive-distillation flowsheet in *CO2-from-NG: extractive distillation, a hard mixture and a cold-start caveat*: solved cold, HEOS's density solver fails at an extreme trial point during the initial-guess search; solved from the flowsheet's saved state (or any warm start), it converges cleanly with no errors — the same physical problem, the same converged answer, the only difference is whether the solver ever has to evaluate that one extreme point.

Practical guidance, confirmed on this flowsheet: for a wide-boiling-range or otherwise numerically

hard mixture (light components alongside a much heavier one, near an azeotrope, or any flowsheet that fails only on a cold/Reset solve), do an initial solve with the **same** CoolProp property package set to a **cubic backend** (PR or SRK — fast and robust across a wide composition range) to get the flowsheet to a converged state, then, on that CoolProp package instance, switch the EOS selection to **HEOS** and re-solve from the now warm-started state. Since the EOS selection is a single `Edit()` setting on one package instance (*Equation-of-state / backend selection*), no flowsheet restructuring or second property package is needed — reset -> solve under CoolProp/cubic -> switch to CoolProp/HEOS -> solve. Verified end-to-end on *CO2-from-NG: extractive distillation, a hard mixture and a cold-start caveat*: this exact sequence converges cleanly where a cold HEOS solve does not.

3.5 PH/PS flash strategy

The state-function (PH/PS) flash — find the state matching a target enthalpy or entropy at fixed pressure — is the one flash CoolProp does not solve as a single robust native call across the package's whole operating history (see *Flash and Property Routing* for the detailed mechanics and the historical native-flash failure mode this motivated). The package therefore offers **two interchangeable strategies**, selectable in the `Edit()` GUI next to the EOS dropdown and persisted with the flowsheet:

Strategy	Behaviour
C# (default)	A quality-based flash: bracket the target between the bubble and dew properties via CoolProp's saturation (`PQ`) flash, and if the target lies inside that bracket, invert the smooth, metastable-free $H(Q)$/$S(Q)$ relation directly on quality; otherwise solve on temperature. Every result is verified against the target before being returned. This is the strategy validated end-to-end on all three reference flowsheets (*Validation*).
Native	CoolProp's own mixture `HSU_P` (PH) / `PSmolar` (PS) flash, likewise verified against the target — falling back to the C# strategy, per flash, if the native call throws or the verification fails. With the compound-order fix in place (above), the native flash now converges reliably on the validation flowsheets (zero fallbacks observed on TEALARC), a material change from earlier CoolProp releases where it would occasionally converge silently to the wrong temperature. It can still converge to a measurably different result at a marginal (near-zero-approach) operating point than the C# strategy does.

Both strategies are always available without a rebuild — the choice is a configuration setting, not a compile-time one — so a flowsheet that behaves better under one strategy can switch without code changes, and the setting travels with the flowsheet file. An environment variable (`COOLPROPCO_PHPS=native` / `=csharp`) can override the configured strategy for diagnostics (e.g. CI runs, A/B benchmarking) without touching the saved flowsheet state.

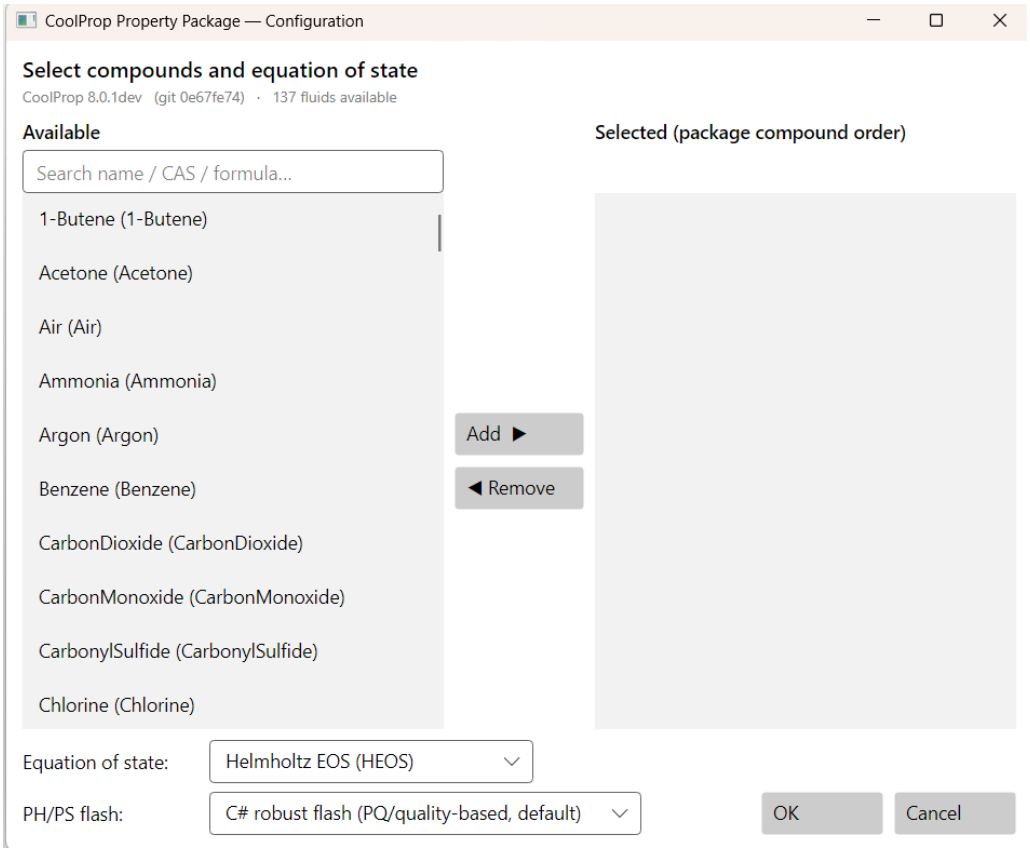


Figure 3.1: The PH/PS flash strategy control, alongside the EOS selector, in the `Edit()` configuration dialog (shown here defaulted to the C# strategy).

Chapter 4

Flash and Property Routing

4.1 Flash routing overview

`CalcEquilibrium` dispatches on the pair of specifications the PME supplies (*Introduction*, CO 1.1 §7). Four of the five flash types route directly to a native CoolProp input pair on the equilibrium handle (*The AbstractState handle model*); the fifth — the state-function flash — is handled by dedicated logic described below.

Flash	CoolProp input pair	Notes
TP	`PT_INPUTS`	Direct. Wrapped in the micro-perturbation retry (below).
PVF	`PQ_INPUTS`	Bubble/dew (saturation) flash at fixed pressure. Direct; also perturbation-wrapped.
TVF	`QT_INPUTS`	Bubble/dew flash at fixed temperature. Direct; also perturbation-wrapped.
PH	see below	*The C# PH/PS flash* (default) or *The native PH/PS flash (opt-in)* (*PH/PS flash strategy*).
PS	see below	Same as PH, on entropy.

4.2 The C# PH/PS flash

CoolProp's `update(PT_INPUTS, ...)` can, at or near a phase boundary, return a **metastable** single-phase density root instead of the stable two-phase result — verified directly: evaluated exactly at the bubble or dew temperature, it can report a single-phase quality ($Q = -1$) rather than the physical boundary value. Bisecting a state-function flash on temperature using such calls as the residual is fragile exactly where it matters most: the bracket endpoints can read single-phase enthalpies for a target that is genuinely two-phase, and the solve clamps to a temperature far from the true answer (observed: a 95% enthalpy error on a mixed-refrigerant loop during early validation).

CoolProp's **saturation** (`PQ`) flash does not have this failure mode, and enthalpy/entropy are smooth, monotone functions of quality along the saturation line at fixed pressure. The C# strategy uses that fact directly:

1. Evaluate the bubble ($Q = 0$) and dew ($Q = 1$) property at the flash pressure via the saturation flash.
2. **If the target lies inside** [bubble, dew], the state is two-phase: invert the monotone $H(Q)$ / $S(Q)$

relation directly on $Q \in [0, 1]$ using the Illinois method (never touches the metastable-prone unimposed PT path).

3. **Otherwise** the state is genuinely single-phase: solve on temperature via *Blind temperature solve (single-phase branch)*, then commit that state with a final PT update (wrapped in the perturbation retry, below).
4. **Verify**: read back the achieved property and compare against the target ($|\text{got} - \text{target}| \leq 10^{-2} + 10^{-3}|\text{target}|$). A miss is logged (non-fatal — failing hard here would abort the PME's solve iteration on what is typically a transient, unreachable recycle guess rather than a real infeasibility) but the flash still returns its best-effort result.

Implementation: `CoolPropEngine.FlashPhPs()`.

4.2.1 Blind temperature solve (single-phase branch)

For the genuinely single-phase branch, H and S are monotone increasing in T at fixed P, so the target temperature is found by root-finding the residual $H(T) - H_{\text{target}}$ (or the entropy analogue), each evaluation being one PT flash.

- **Bracket**: start wide (200–600 K) and walk the endpoints inward past any T where CoolProp cannot evaluate, until both residuals are finite; if they don't straddle zero, widen further (up to 10 attempts) before giving up and returning the closer endpoint (a clamp, not a failure — the caller's verification step in *The C# PH/PS flash* will flag it if that clamp is materially wrong).
- **Root-find**: the **Illinois method** (modified regula falsi — a secant step with endpoint-weight halving to prevent the classic stalled-endpoint problem plain regula falsi has on convex functions). This keeps the robustness of a bracketed method while converging super-linearly on the smooth monotone residual: typically 6–10 evaluations, versus 30–80 for plain bisection on the same bracket.

Implementation: `CoolPropEngine.BlindSolveT()`.

4.3 The native PH/PS flash (opt-in)

CoolProp also exposes a native mixture state-function flash (`HSU_P` for PH; the entropy analogue for PS). Historically this package did not default to it: earlier CoolProp releases could converge it to a **silently wrong** temperature on hard mixtures, which is worse than an outright failure because nothing signals the result is incorrect. That specific class of unreliability is what the upstream mixture-flash improvements referenced in *Introduction* primarily address, and it is also downstream of the compound-order fix in *Compound-order alignment to the Material Object*: a scrambled feed is itself sufficient to make *any* flash — native or not — converge to a plausible-looking wrong answer.

With both fixed, the native flash is offered as a selectable, **verified** strategy (*PH/PS flash strategy*) rather than a blind default switch:

1. Run the native input-pair update (`HmolarP_INPUTS`/`HmassP_INPUTS` for PH, `PSmolar_INPUTS`/`PSmass_INPUTS` for PS) on the equilibrium handle.
2. Read back the achieved property and compare against the target with the same tolerance as the C# strategy's verification step.
3. If the native call throws, or the verification fails, **fall back to the C# strategy for that flash** — the mismatch is logged (`PHNATIVE-FALLBACK`) and the C# path's own result is returned, so a native-flash failure never surfaces as a PME-visible error.

Because the fallback is per-flash and silent to the caller, the native strategy can be selected with no correctness risk beyond "falls back to the proven default" — the only externally visible difference, when

the native flash *does* converge, is that it can land on a measurably different temperature than the C# path at a marginal (near-zero-approach) operating point, as noted in *PH/PS flash strategy*.

Implementation: `CoolPropEngine.TryNativePhPs()`.

4.4 Micro-perturbation convergence retry

Independent of which flash type is running, CoolProp's mixture successive- substitution / density solve can fail to converge at a point a hair from a solution — not because the state is infeasible, but because the iteration itself stalls (`"PT flash lost a phase density solve during successive substitution"`, `"solver_rho_Tp ..."`). This is most visible on a PME's recycle-convergence trial compositions, which probe states a real flowsheet solve never lands on for long, and it is history-dependent: the identical $(T, P, \mathbf{x})$ that fails mid-solve can succeed as a fresh, isolated call, because CoolProp's flash carries internal state across the sequence of calls a PME makes.

Every native `update()` call the engine makes is wrapped with a retry: on a recoverable failure (matched by message substring — "density solve", "successive substitution", "solver_rho"), the continuous spec variable is nudged by a small, escalating fraction and the update retried:

$$f \in \{\pm 3 \times 10^{-5},\ \pm 10^{-4},\ \pm 3 \times 10^{-4},\ \pm 10^{-3},\ \pm 3 \times 10^{-3}\}$$

— perturbing P for a `PQ` flash, T otherwise. The first nudge that converges wins; the resulting state differs from the requested one by at most 0.3%, an offset that washes out as the PME iterates the recycle to convergence. If no nudge converges, the original exception is re-thrown unchanged — a genuinely infeasible specification still fails loudly rather than silently returning a nearby-but-wrong state.

Implementation: `CoolPropEngine.UpdateResilient()`, `IsRecoverableFlashFailure()`, `PerturbFractions`. Recoveries are counted (`PerturbationRecoveries`) and traced through the CAPE-OPEN logger for diagnostics.

Chapter 5

Transport-Property Fallbacks

CoolProp's HEOS backend has native, fluid-specific viscosity and thermal conductivity models for most of its pure fluids and their mixtures. The **cubic** backends (PR, SRK) do not: they solve density and phase equilibrium from the cubic EOS but carry no transport-property correlations of their own, so a native transport query on a cubic-backend handle simply has nothing to return. A package that only wraps CoolProp's native transport output would leave viscosity and thermal conductivity unavailable to the PME whenever the user selects a cubic EOS.

To keep the property package usable across all three EOS choices, every transport-property request follows a **try-native-first, fall back on failure** strategy:

1. Query CoolProp's native `viscosity` / `conductivity` output on the current `AbstractState` handle.
2. If CoolProp has no model for the fluid/backend combination (the native call fails), fall back to a correlation from `TransportFallback` (`CoolPropCO.Core/Transport/TransportFallback.cs`) — a set of corresponding-states methods ported from the `ThermoCalc.CapeOpen` reference package's `ViscosityCalc` service.

This means: on HEOS the package normally returns CoolProp's own transport values; on PR/SRK (and on any HEOS fluid combination CoolProp itself does not cover) it transparently returns the fallback correlation instead, so the PME always gets a usable number rather than a hard property failure. This chapter documents those fallback correlations, all implemented in `TransportFallback.cs`.

Note

The fallback set here is intentionally smaller than the ThermoCalc reference it was ported from: **water-specific correlations** (DIPPR viscosity/conductivity, IAPWS surface tension, Meybodi hydrocarbon-water interfacial tension), the **Letsou-Stiel** liquid-viscosity alternative, the **Stiel-Thodos high-pressure conductivity correction**, and **Kesler-Lee pseudo-component estimation** are all present in ThermoCalc but not ported here — CoolProp's own fluid database already covers water and the package does not (yet) support user-defined pseudo-components (`Component.IsPseudo` is reserved for a future release; see *Design Decisions*). Only the correlations actually needed to cover the cubic-backend gap are implemented.

5.1 Gas Viscosity

The Lee-Gonzalez-Eakin (1966) correlation [6] is used for gas-phase dynamic viscosity. It is a density-corrected exponential model valid for natural-gas-like systems at moderate pressures.

The gas density is computed from the compressibility factor:

$$\rho = \frac{P \cdot 10^5 \, M_w}{Z \cdot 8314 \, T} \cdot 10^{-3} \quad [\text{g/cm}^3] \tag{5.1}$$

where P is pressure [bar], M_w is mixture molecular weight [g/mol], Z is the compressibility factor (from the molar density already computed by CoolProp for the state, so no extra EOS solve is needed), and T is temperature [K]. Temperature converts to Rankine: $T_R = 1.8\,T$.

Three intermediate parameters are computed:

$$\begin{aligned} A &= \frac{(9.4 + 0.02\,M_w)\,T_R^{1.5}}{(209 + 19\,M_w + T_R) \cdot 10^4} \\ B &= 3.5 + \frac{986}{T_R} + 0.01\,M_w \\ C &= 2.4 - 0.2\,B \end{aligned} \tag{5.2}$$

The dynamic viscosity [cP], converted to Pa·s for CAPE-OPEN output:

$$\mu = A \exp(B\,\rho^C) \tag{5.3}$$

Implementation: `TransportFallback.TryViscosity()` -> `LeeGonzalez()`

5.2 Liquid Viscosity

5.2.1 Lohrenz-Bray-Clark (1964)

The LBC method [7] computes liquid viscosity as a residual contribution added to a dilute-gas base, following the formulation in [8] (Eqs. 10.37–10.41). This is the package's only liquid-viscosity fallback (ThermoCalc's Letsou-Stiel alternative is not ported — see the note above).

Dilute-gas viscosity is computed per component using the corresponding-states parameter:

$$\xi_i = \frac{T_{c,i}^{1/6}}{\sqrt{M_{w,i}}\,P_{c,i}^{2/3}} \tag{5.4}$$

where $P_{c,i}$ is in atmospheres. The per-component dilute viscosity depends on reduced temperature $T_r = T/T_{c,i}$:

$$\mu_i^\circ = \begin{cases} 0.00034\, T_r^{0.94}/\xi_i & T_r < 1.5 \\ 0.0001778\,(4.58\, T_r - 1.67)^{5/8}/\xi_i & T_r \geq 1.5 \end{cases} \tag{5.5}$$

The mixture dilute viscosity uses Herning-Zipperer [9] weighting:

$$\mu_{\text{mix}}^\circ = \frac{\sum_i x_i\, \mu_i^\circ \sqrt{M_{w,i}}}{\sum_i x_i \sqrt{M_{w,i}}} \tag{5.6}$$

Residual viscosity is obtained from a fourth-order polynomial in reduced density. The mixture inverse-viscosity parameter is:

$$\psi = \frac{\left(\sum_i x_i T_{c,i}\right)^{1/6}}{\sqrt{\sum_i x_i M_{w,i}}\ \left(\sum_i x_i P_{c,i}\right)^{2/3}} \tag{5.7}$$

The reduced density is $\rho_r = \rho_m/\rho_c$, where the mixture molar density ρ_m is taken directly from CoolProp's converged state (again, no separate EOS evaluation) and $\rho_c = 1/\sum_i x_i V_{c,i}$. The component critical volume $V_{c,i}$ uses CoolProp's reported V_c when available, falling back to the Riedel correlation $Z_c = 0.2905 - 0.085\,\omega$ otherwise (see *Design Decisions* for where compound constants come from).

The LBC polynomial is:

$$\mathcal{P}(\rho_r) = a_1 + a_2\,\rho_r + a_3\,\rho_r^2 + a_4\,\rho_r^3 + a_5\,\rho_r^4 \tag{5.8}$$

with coefficients:

Coefficient	Value
a_1	0.1023
a_2	0.023364
a_3	0.058533
a_4	-0.040758
a_5	0.0093324

The liquid viscosity [cP]:

$$\mu_L = \frac{\mathcal{P}^4 - 10^{-4}}{\psi} + \mu_{\text{mix}}^\circ \tag{5.9}$$

with a clamp to μ_{mix}° if the residual term goes negative (possible at low reduced density, outside the correlation's intended range). Converted to Pa·s for CAPE-OPEN output.

Implementation: `TransportFallback.TryViscosity()` -> `Lbc()`

5.3 Thermal Conductivity (Gharagheizi 2013)

Both gas- and liquid-phase thermal conductivity use the Gharagheizi [10] quantitative structure-property relationship (QSPR) method — a corresponding-states method requiring only T_b, M_w, P_c, and ω per component. Unlike ThermoCalc, no high-pressure (Stiel-Thodos) correction is applied — the fallback is used only on the cubic backends, where the priority is a physically reasonable low-to-moderate-pressure estimate rather than exact high-pressure behaviour.

Gas mixture conductivity uses Wassiljewa-Herning-Zipperer mixing:

$$k_{\text{mix}} = \frac{\sum_i y_i \, k_i \sqrt{M_{w,i}}}{\sum_i y_i \sqrt{M_{w,i}}} \tag{5.10}$$

Liquid mixture conductivity uses simple linear mole-fraction mixing:

$$k_{\text{mix}} = \sum_i x_i \, k_i \tag{5.11}$$

Implementation: `TransportFallback.TryThermalConductivity() -> GharagheiziGas() / GharagheiziLiquid()`

5.4 Surface Tension

5.4.1 Brock-Bird (1955)

Two-phase surface tension (*Design Decisions* — a `CalcTwoPhaseProp` request, not single-phase) is computed per component using the Brock-Bird [11] corresponding-states method, evaluated on the liquid-phase composition:

$$\begin{aligned} Q &= 0.1196 \left(1 + \frac{T_{b,r} \ln(P_c/1.01325)}{1 - T_{b,r}}\right) - 0.279 \\ \sigma_i &= P_c^{2/3} \, T_c^{1/3} \, Q \, (1 - T_r)^{11/9} \times 10^{-3} \end{aligned} \tag{5.12}$$

where P_c is in bar, $T_{b,r} = T_b/T_c$ is the reduced normal boiling point, and $T_r = T/T_c$. Components at or above their critical temperature (no meaningful surface tension) are skipped. Mixture surface tension uses linear mole-fraction mixing over the liquid composition:

$$\sigma_{\text{mix}} = \sum_i x_i \, \sigma_i$$

The result is in N/m. This is the package's **only** surface-tension model — CoolProp's own two-phase surface tension is used when available (see `SurfaceTension()` in `CoolPropEngine.cs`, evaluated on the dedicated liquid handle — *Architecture*); Brock-Bird is the fallback when it is not.

Implementation: `TransportFallback.TrySurfaceTension()` -> `BrockBirdPure()`

5.5 Component normal boiling point

All three fallback correlations need each component's normal boiling point T_b. When the compound catalog does not supply it directly, it is estimated from the Edmister relation using T_c, P_c, and ω:

$$\begin{aligned} k &= \frac{7}{3} \frac{\omega + 1}{\log_{10}(P_c/P_{\text{atm}})} \\ T_{b,r} &= \frac{k}{1+k}, \qquad T_b = T_c \, T_{b,r} \end{aligned} \tag{5.13}$$

Implementation: `TransportFallback.Tb()`

Chapter 6

The Incompressible Companion Package

Utility and heat-transfer-fluid streams (glycol/water, brines, thermal oils) do not need — and cannot use — a VLE flash: CoolProp's `INCOMP` backend models them as single-phase liquids fitted directly to density, heat capacity, viscosity, and conductivity correlations, with no equation of state and no vapour phase. Rather than stretch the mixture property package to cover this case, it is a **separate** CAPE-OPEN 1.1 property package (`CoolPropIncompressiblePackage`, CLSID `837C291D-A3C5-4E4E-A981-448A6F0896FA`, ProgId `CoolPropCO.IncompressiblePackage`) purpose-built around what CoolProp's `INCOMP` backend actually provides.

6.1 Scope

One package instance wraps **one** incompressible fluid: either a pure incompressible or one binary solution at a configurable concentration (e.g. propylene glycol/water at a chosen mass fraction). It is liquid-only — one phase, no fugacity, no K-values — reported to the PME as a single pseudo-compound (the fluid itself has no CAS number to report).

CoolProp's incompressible catalogue is loaded from an embedded copy of its fluid-metadata JSON: 126 fluids (74 pure, 52 solutions), each carrying description, literature reference, concentration basis (mass / volume / mole fraction), valid temperature range, and valid concentration range. A solution's concentration is bounded to its fitted range in the `Edit()` GUI (e.g. MEG 0–0.6, aqueous LiBr 0–0.75) — the underlying correlations are not extrapolated outside where they were fitted.

6.2 Why mass basis only

CoolProp's `INCOMP` backend exposes **only mass-basis** properties (density, enthalpy, entropy, heat capacity, viscosity, conductivity); molar quantities — molar mass, molar density/enthalpy/entropy, speed of sound — are not available and error if requested. The package follows this exactly: every property it advertises is on a mass basis.

CAPE-OPEN still expects a package to be able to report a molecular weight (for PME-side mole<->mass bookkeeping on streams that mix compressible and incompressible packages). Since the fluid has no real molar mass, the package reports a **nominal** value (18.015 kg/kmol — water's, chosen only because it's a familiar, unremarkable number). This is safe specifically *because* every property the package reports is mass-basis: the nominal value cancels out of any mole<->mass round-trip the PME performs ($n \cdot h_{\text{molar}} = (m/M_w)(h_{\text{mass}} \cdot M_w) = m \cdot h_{\text{mass}}$) rather than introducing an error.

6.3 Low-level construction

The `AbstractState` is created via `factory("INCOMP", name)`; for a solution, the concentration is set afterward via `set_fractions([x])` — **not** by encoding it into the fluid name string (`"Name[0.25]"`). That bracketed form is accepted by CoolProp's high-level `PropsSI` convenience function but is rejected by the low-level `factory()` call the package uses throughout (*Architecture*), so concentration is always set as a separate step.

6.4 Flashes supported

Only TP, PH, and PS are meaningful for a single incompressible liquid phase (no vapour fraction to specify against):

Flash	CoolProp call	Notes
TP	*(none needed)*	Trivial — T and P are already the two independent variables for an incompressible liquid.
PH	`HmassP_INPUTS`	Mass-basis enthalpy target.
PS	`PSmass_INPUTS`	Mass-basis entropy target.

Requesting PVF/TVF (vapour-fraction) flashes against an incompressible fluid is a modelling error on the flowsheet side, not a state the package attempts to solve — it raises a clear error rather than silently returning a physically meaningless result.

6.5 Freezing point

A solution's freezing point is concentration-dependent, and is exposed as a CoolProp `T_freeze` query on the configured (fluid, concentration) pair — reported in the `Edit()` GUI live as the concentration is adjusted, so a user configuring a glycol mixture can see immediately whether a chosen concentration is adequate for the coldest temperature the stream will see.

> **Note**
>
> Not every fluid in the catalogue has a fitted freezing-point correlation in CoolProp's own data — this is a data-availability gap in CoolProp itself, not a package limitation. For a fluid without one (e.g. MIT Seawater, aqueous LiBr), the query does not raise an error or return NaN: it silently evaluates to 0 K (displayed as `0.00 K (-273.1 °C)` in the GUI — this is what the figure below actually shows for MIT Seawater). **Only a non-zero (non-absolute-zero) freezing point is a real, fitted value** — a reported 0 K means no correlation is available for that fluid, not that it never freezes.

6.6 Configuration GUI

`IncompressibleEditorWindow` lists the catalogue with a search box, a concentration control (a bounded numeric up/down, shown only for solutions and clamped to the fluid's fitted $[x_{\min}, x_{\max}]$ range), and live display of the fluid's description, literature origin, basis, valid temperature range, valid concentration range, and current freezing point. A button opens the fluid's pre-generated fitting-quality report (a PDF hosted on coolprop.org) directly from the dialog, so the correlation's accuracy and validity range can be checked against the source without leaving the GUI.

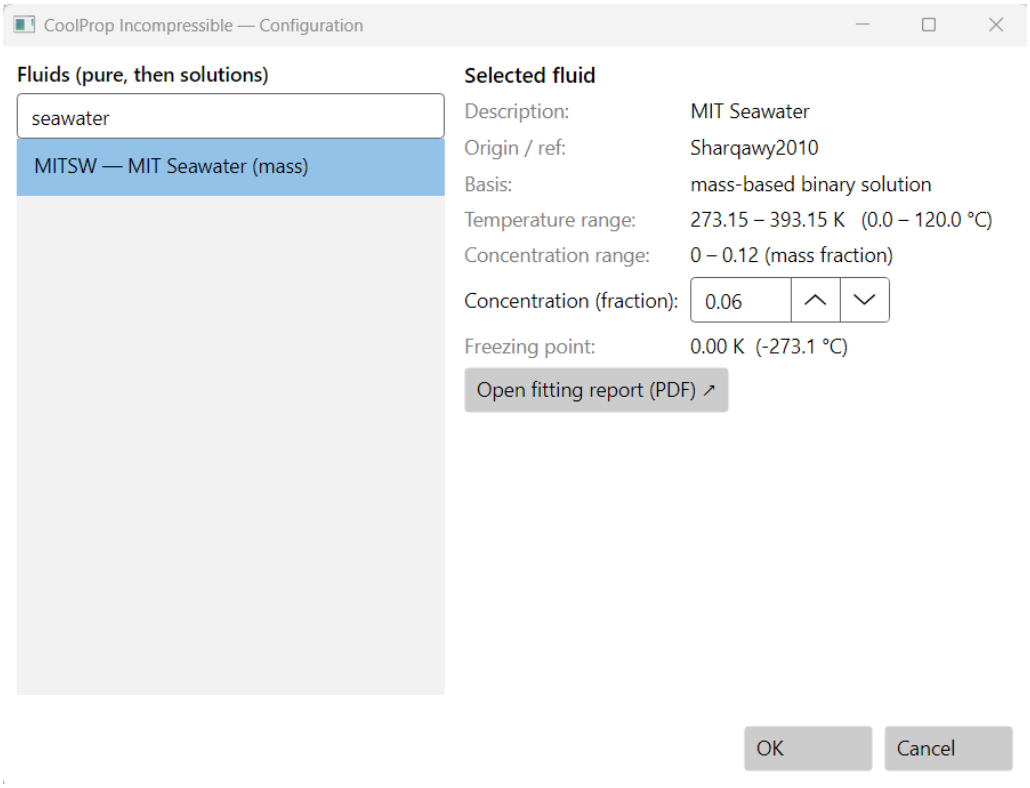


Figure 6.1: The incompressible-fluid selection / concentration dialog shown by `Edit()` on the companion package, here configuring MIT Seawater ("Sharqawy2010" per CoolProp's own origin/reference tag) at 6% mass concentration. The displayed freezing point, `0.00 K (-273.1 °C)`, is the no-correlation-available case explained above, not a real result.

Part III

Validation

Chapter 7

Validation

The package is validated end-to-end, the way it is actually used: six complete process flowsheets, solved in COFE, comparing the CoolProp-backed property package against a native reference package on the *same* flowsheet topology. This is a deliberately different kind of validation than checking CoolProp's own equations of state against experimental data — that validation is CoolProp's own, upstream (*Introduction*) and is not repeated here. What is validated here is that **the CAPE-OPEN wrapper delivers CoolProp's thermodynamics correctly into a real, converged flowsheet solve**, across single-component, hydrocarbon-mixture, hard-mixture-distillation, and multi-stage-refrigeration cases — which is precisely where an interface bug like the compound-order issue in *Design Decisions* shows up, and an EOS-accuracy check alone would never catch it.

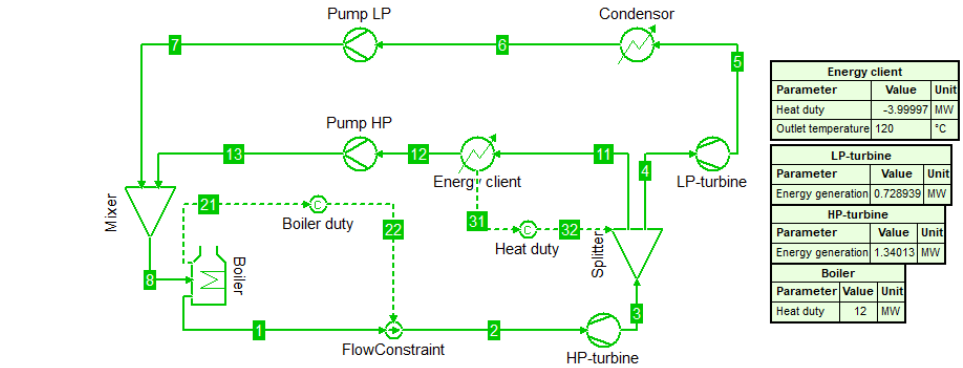
Four of the six flowsheets (CHP, TEALARC, Snøhvit, ASU) are solved with COFEStand's headless solver from a full reset (`COFEStand.exe /r <flowsheet.fsd>`). The remaining two (CO2-from-NG and the CO_2 compression/liquefaction train) are solved from the flowsheet's saved state instead — note that `/r` means *reset before solve*, so omitting it is what gives the warm start normal interactive COFE use provides. For CO2-from-NG the reset case is documented separately as a known, worked-around limitation in *CO2-from-NG: extractive distillation, a hard mixture and a cold-start caveat*. The comparison scripts read the solved `.fsd` archives directly (no simulation re-run at doc-build time) and are committed alongside their vendored CSV output under `flowsheets/` and `figures/data/` for reproducibility.

7.1 CHP: pure water, single-component routing

A combined-heat-and-power steam-Rankine cycle: a boiler raises steam to 653 K / 22 bar, an HP turbine and LP turbine (extraction, isentropic efficiency specified) generate shaft power, an "Energy client" HeaterCooler extracts process heat at 393 K (the CHP heat offtake), and the condensate is returned to the boiler through two feedwater pumps. Every stream is **pure water** — the exact degenerate composition the package routes to a dedicated pure-fluid CoolProp handle rather than the mixture backend (*Single-component (pure) compositions*). This flowsheet is the validation of that routing decision specifically.

Table 7.1: CHP: native vs. CoolProp (pure-fluid routing)

Metric	Native	CoolProp	Difference
Net shaft power [kW]	2057.1	2057.1	$< 0.01\%$
Boiler duty [kW]	12000.0	12000.0	$< 0.01\%$
CHP heat (client) [kW]	4000.0	4000.0	$< 0.01\%$
Power efficiency	17.142 %	17.143 %	< 0.01 pt
CHP (power+heat) efficiency	50.475 %	50.476 %	< 0.01 pt



Energy client		
Parameter	Value	Unit
Heat duty	-3.99997	MW
Outlet temperature	120	°C

LP-turbine		
Parameter	Value	Unit
Energy generation	0.728939	MW

HP-turbine		
Parameter	Value	Unit
Energy generation	1.34013	MW

Boiler		
Parameter	Value	Unit
Heat duty	12	MW

Stream	1	3	5	6	8	Unit
Pressure	22	2	0.1	0.1	22	bar
Temperature	380	203.294	45.8075	40	72.9334	°C
Flow rate	14.9301	14.9301	8.8635	8.8635	14.9301	ton / h
Enthalpy	13440.7	7619.28	2285.55	-41199.8	-38686.3	J / mol
Mole frac Water	1	1	1	1	1	
Vapor phase						
phase fraction	1	1	0.998936			
Liquid phase						
phase fraction			0.00106402	1	1	

Title:	A simple CHP cycle
Author:	Torbjørn Pettersen
Affiliation:	Thermos AS
Created:	Apr 27, 2009

Figure 7.1: The CHP steam-Rankine cycle flowsheet in COFE (original flowsheet by Torbjørn Pettersen, Thermos AS, 2009), with the stream table and turbine/boiler/client duty summary as solved.

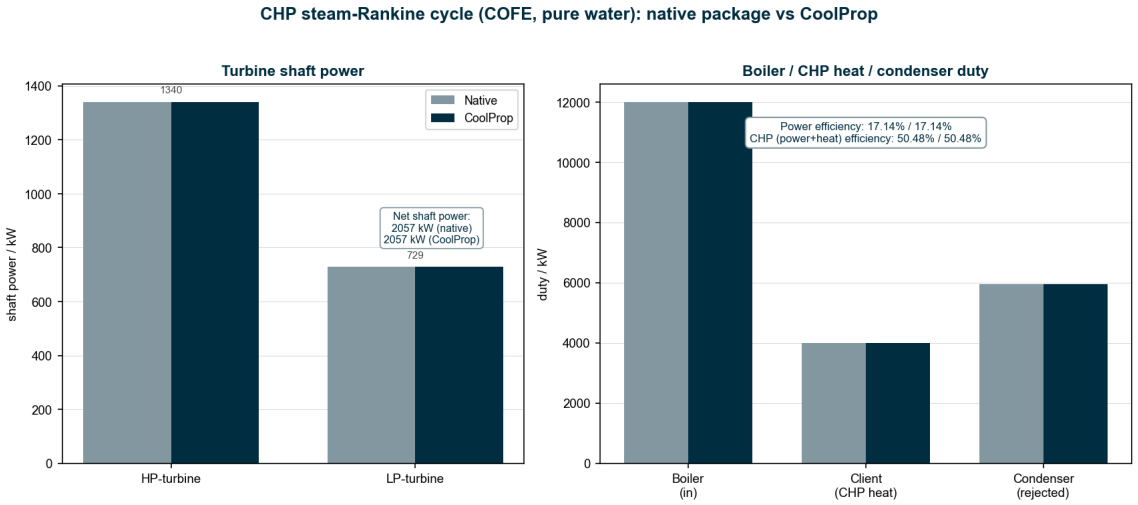


Figure 7.2: Turbine shaft power, boiler/CHP-heat/condenser duty, and overall efficiency: native water package vs CoolProp (pure-fluid routing).

Agreement is better than 0.1% on every metric — as expected for pure water, where both packages are ultimately evaluating essentially the same correlation (CoolProp's own high-accuracy water EOS, IAPWS-95-class, via the pure-fluid handle). This is the clean baseline case: no mixture flash, no compound-order question, nothing for the interface layer to get wrong.

7.2 TEALARC: dual mixed refrigerant, the C2/C3 case study

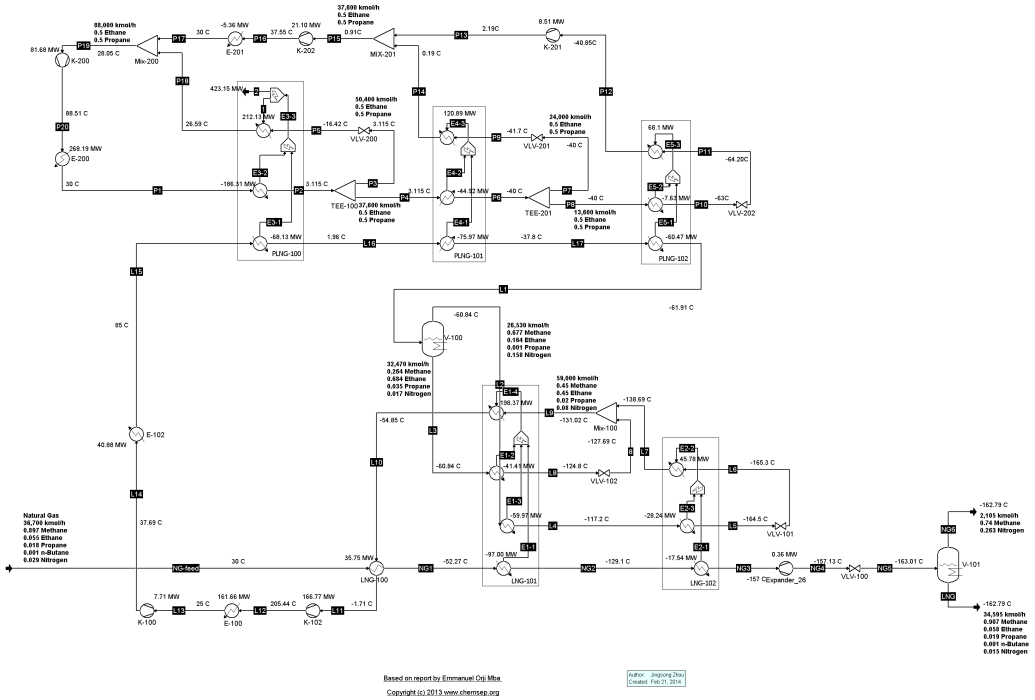


Figure 7.3: The TEALARC dual mixed-refrigerant LNG liquefaction flowsheet in COFE.

A natural gas feed (C1/C2/C3/nC4/N2, 40 bar, 303 K) is liquefied and sub-cooled to LNG (~109 K, 1.1 bar) against two mixed-refrigerant cycles — a 5-compressor train (K-100, K-102, K-200, K-201, K-202), each specifying a pressure ratio and isentropic efficiency rather than a fixed duty, so shaft power is genuinely EOS-computed from the enthalpy rise across each compression stage. This is exactly the flowsheet — and stream P1, the C2/C3 precool loop condenser, is exactly the state — described in *Compound-order alignment to the Material Object*: a 50/50 ethane/propane condenser at 30 °C, 22.5 bar that a compound-order mismatch had previously made read as all-liquid instead of two-phase.

Table 7.2: TEALARC: native cubic vs. CoolProp HEOS

Metric	Native	CoolProp	Difference
Total compressor power [MW]	283.78	278.21	-2.0%
P1 vapour fraction Q	0.443	0.398	both two-phase
LNG recovery [% feed mass]	93.78	93.64	-0.14 pt
Specific power [kWh/t LNG]	465.6	457.1	-1.8%

P1's vapour fraction — the specific stream at the center of the compound-order investigation — is genuinely two-phase in **both** packages, with the cubic and HEOS quality differing by an amount consistent with the two EOS's known behaviour on this system, not with a scrambled feed. The 5-compressor power total differs by 2.0%, and the figure above shows each individual machine's power tracking that total-power difference closely rather than showing an outlier — the signature of a consistent EOS differ-

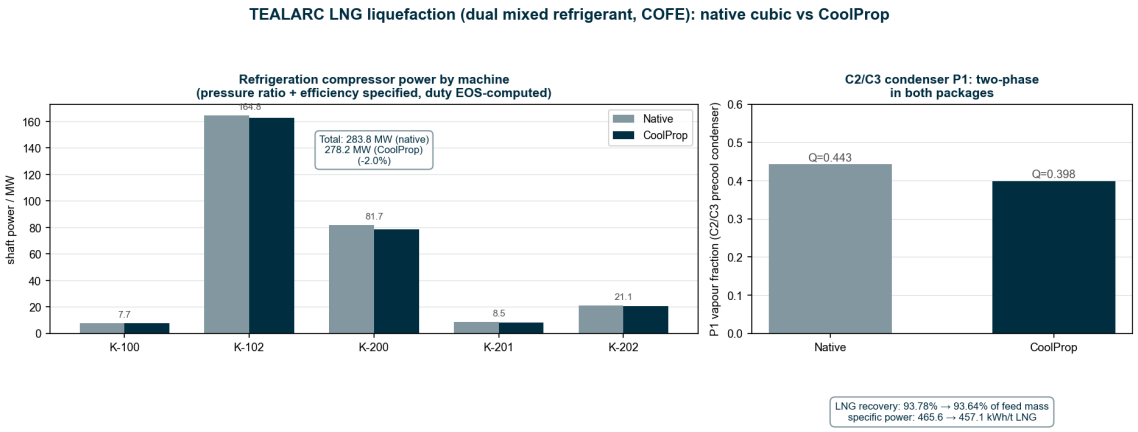


Figure 7.4: Compressor shaft power by machine, the C2/C3 condenser P1 vapour fraction, and LNG recovery: native cubic vs CoolProp (HEOS), fixed compound-order alignment.

ence along the compression path, not an isolated flash error on one machine.

7.3 CO2-from-NG: extractive distillation, a hard mixture and a cold-start caveat

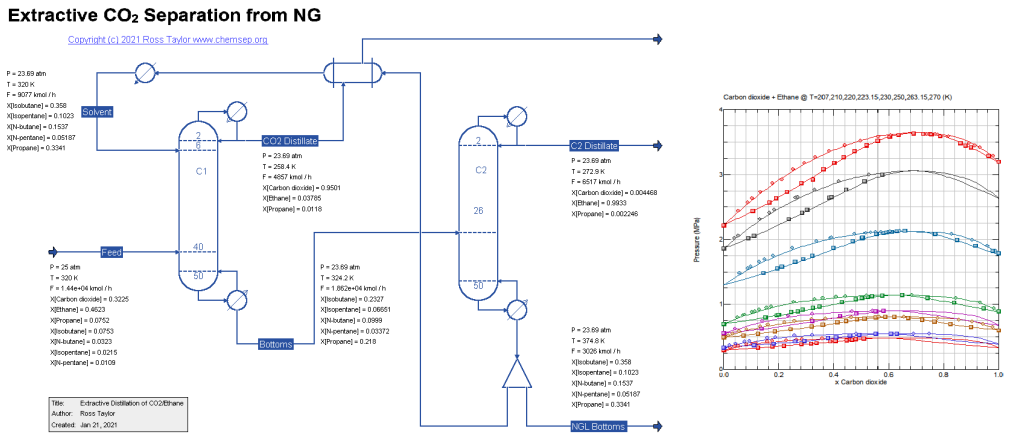


Figure 7.5: The CO2-from-NG extractive-distillation flowsheet (IECR 52, p.10780) in COFE.

CO2 and ethane form a low-boiling azeotrope. Column C1 breaks it by extractive distillation with an added heavy (C3–C5) hydrocarbon solvent, producing a CO2 distillate and a bottoms of ethane + solvent; column C2 recovers the ethane overhead and recycles the solvent. The feed spans a much wider boiling range than TEALARC or Snøhvit's refrigerant loops — light CO2 and ethane alongside a C3–C5 solvent — and the CO2/ethane relative volatility near the azeotrope is itself EOS-sensitive, making this a genuinely harder mixture-flash case, run through ChemSep columns rather than simple flash drums.

Warning

Cold-start caveat, and the fix. Solved from a fully cold start (a Reset that discards the flowsheet's saved converged state, forcing a full initial-guess search), the CoolProp/HEOS package's mixture density solver fails on an extreme trial composition that search probes. Solved from a warm state — either the flowsheet's saved state, or (confirmed, reproducible) a **reset -> solve with the same**

CoolProp package set to a cubic backend (PR/SRK) -> switch to HEOS -> solve sequence — it converges cleanly with no errors or warnings. This is exactly the scenario documented in *Cold-start initialization on hard mixtures*, and this flowsheet is the worked example: both `.fsd` files here were solved warm (opened and solved from the saved state, as in normal COFE use — matching what a user actually does), not from a COFEStand batch reset.

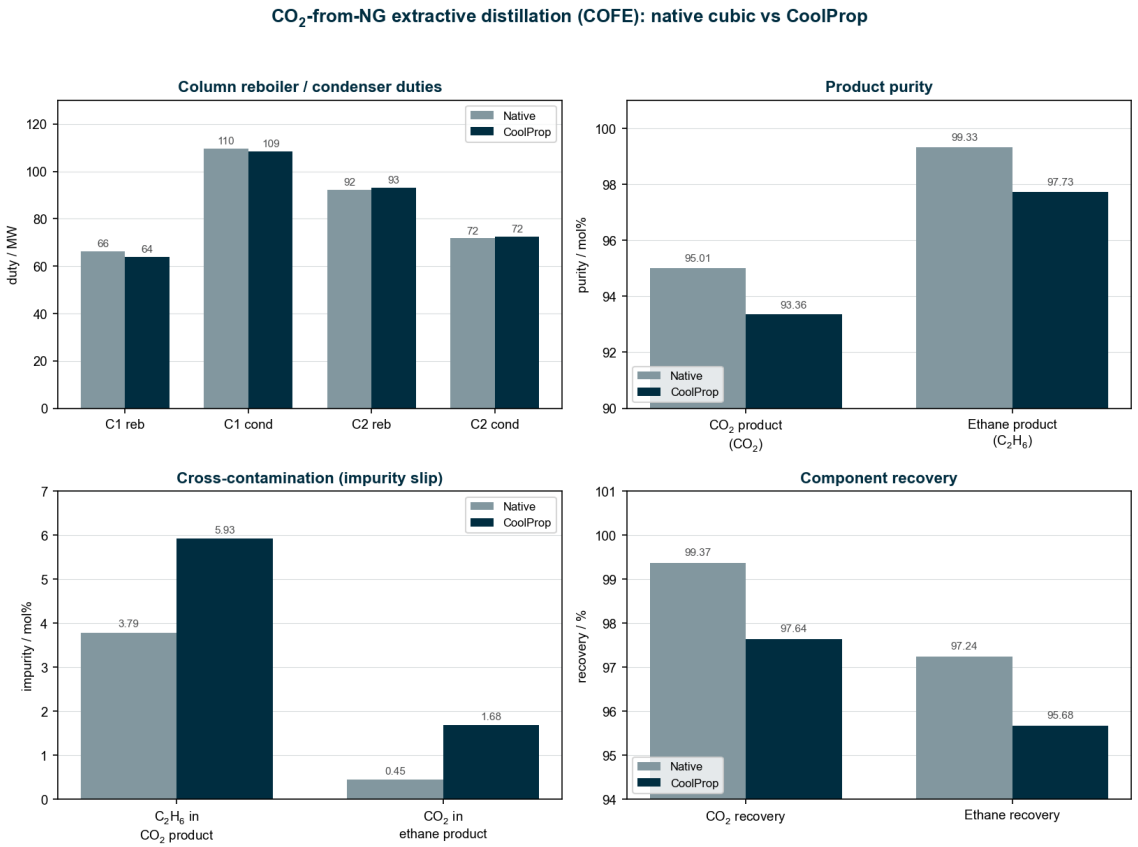


Figure 7.6: Column duties, product purity, cross-contamination, and recovery: native cubic vs CoolProp (HEOS).

Table 7.3: CO2-from-NG: native cubic vs. CoolProp HEOS

Metric	Native	CoolProp	Difference
Total reboiler duty [MW]	158.6	157.0	−1.0%
CO2 product purity [mol%]	95.01	93.36	−1.65 pt
Ethane product purity [mol%]	99.33	97.73	−1.60 pt
CO2 recovery [%]	99.37	97.64	−1.73 pt
Ethane recovery [%]	97.24	95.68	−1.56 pt

Column duties — set almost entirely by the specified reflux ratio and flows, not by the VLE itself — agree to about 1%, consistent with TEALARC/Snøhvit's compression-power agreement. The **product purities and recoveries diverge more** than either refrigeration case (1.5–1.7 percentage points, versus fractions of a percent for CHP or ~2% for TEALARC's compression power) — expected, since this separation depends directly on the CO2/ethane relative volatility right at an azeotrope, the single most EOS-sensitive quantity in this flowsheet, rather than on bulk compression or heat-exchange duty. This is

the validation flowsheet where the choice of EOS is expected to matter most, and it does — in the specific place the underlying chemistry says it should.

7.4 Snøhvit: single mixed refrigerant, multi-stage compression

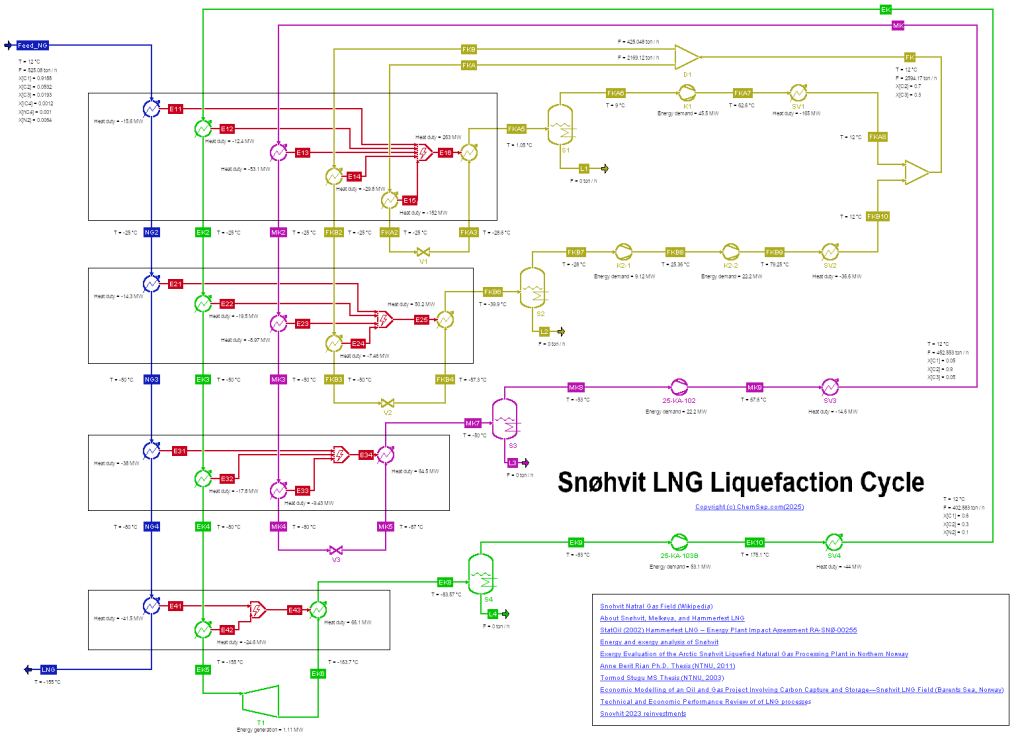


Figure 7.7: The Snøhvit single mixed-refrigerant LNG liquefaction flowsheet in COFE.

A natural gas feed (60 bar, 285.15 K, 8333.33 mol/s) is liquefied to LNG (118.15 K, 145.86 kg/s) against a single mixed-refrigerant cycle compressed across five stages (K1, K2-1, K2-2, 25-KA-102, 25-KA-103B) with a recompression turbine (T1) recovering work from a let-down. Feed and LNG product T/P/flow are confirmed **identical** specifications in both files — this is not assumed, it is read back from both solved flowsheets directly.

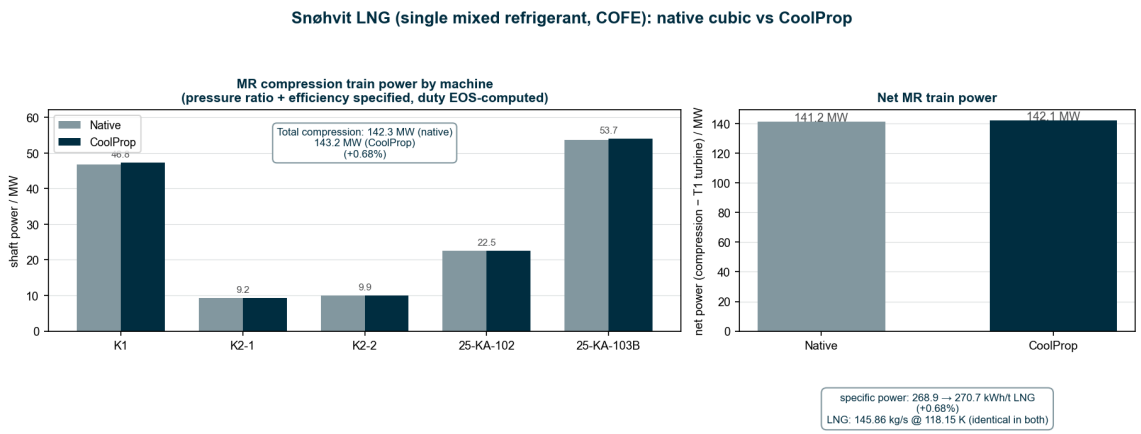


Figure 7.8: MR compression-train shaft power by machine and net train power: native cubic vs CoolProp (HEOS).

Note

One compound-set difference, disclosed rather than hidden. The native flowsheet carries a trace water compound (8 compounds: C1/C2/C3/nC4/iC4/N2/CO2/H2O); the CoolProp flowsheet's feed omits it (7 compounds). This predates the validation exercise and is a defensible modelling simplification in its own right — water is normally removed upstream of the liquefaction section by molecular-sieve dehydration in the real process, so a dehydrated feed is a reasonable compound set for the liquefaction-section property package to carry — but it does mean this comparison is not bit-for-bit identical the way TEALARC's is. It is flagged here explicitly rather than presented as a clean match it is not.

Table 7.4: Snøhvit: native cubic vs. CoolProp HEOS

Metric	Native	CoolProp	Difference
Total compression [MW]	142.28	143.25	+0.68%
Net train power [MW]	141.17	142.14	+0.68%
Specific power [kWh/t LNG]	268.9	270.7	+0.68%

Every machine in the five-stage train and the recompression turbine agree to within about 1% (figure above), and the net-power and specific-power deltas match the total-compression delta almost exactly — again the signature of a consistent, expected EOS difference propagating uniformly through the train, not a localized flash anomaly. Two known, pre-existing, temperature-limited heat exchangers appear in **both** packages' warning output at the flowsheet's coldest multistream-heat-exchange approximation (a discretization limitation of modelling a true multistream exchanger as separate energy-linked Heater-Coolers, present in the native model too) — not a CoolProp-specific artifact.

7.5 Air separation unit (ASU): cryogenic N2/O2/Ar distillation

A 100 kg/s air feed (N2/O2/Ar) is compressed in a three-stage inter-cooled main air-compression train (C1, C2, C3 — each specifying a pressure ratio and isentropic efficiency, so shaft power is EOS-computed from the enthalpy rise across each stage), cooled and liquefied with a turbo-expander providing the refrigeration, and separated in a cryogenic double column plus a crude-argon column into gaseous oxygen (G O2, 98.5 mol% O2), gaseous nitrogen (G N2), liquid argon (L Ar, 99.99 mol% Ar) and a nitrogen-rich

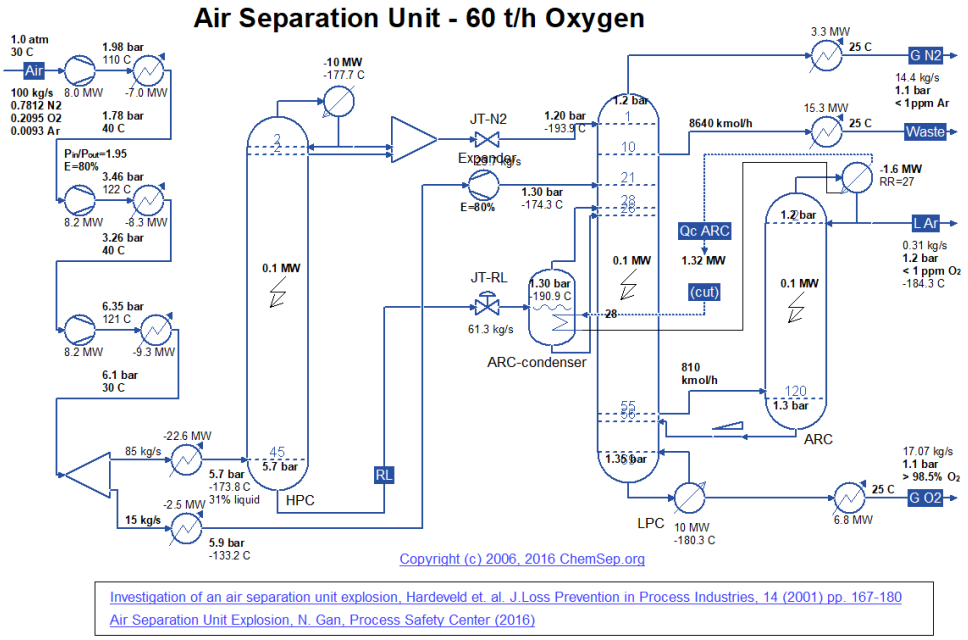


Figure 7.9: The CScasebook cryogenic air-separation flowsheet in COFE (ChemSep case study): three-stage main air compression, turbo-expander, high- and low-pressure columns (HPC/LPC) and the crude-argon column (ARC), producing gaseous oxygen, gaseous nitrogen and liquid argon.

waste stream. This is the low-temperature, tight-boiling-point case: N2, O2 and Ar boil within a few kelvin of one another (~77–90 K), so the column temperatures, relative volatilities and hence the product purities and compression power are a demanding test of the mixture EOS at cryogenic conditions.

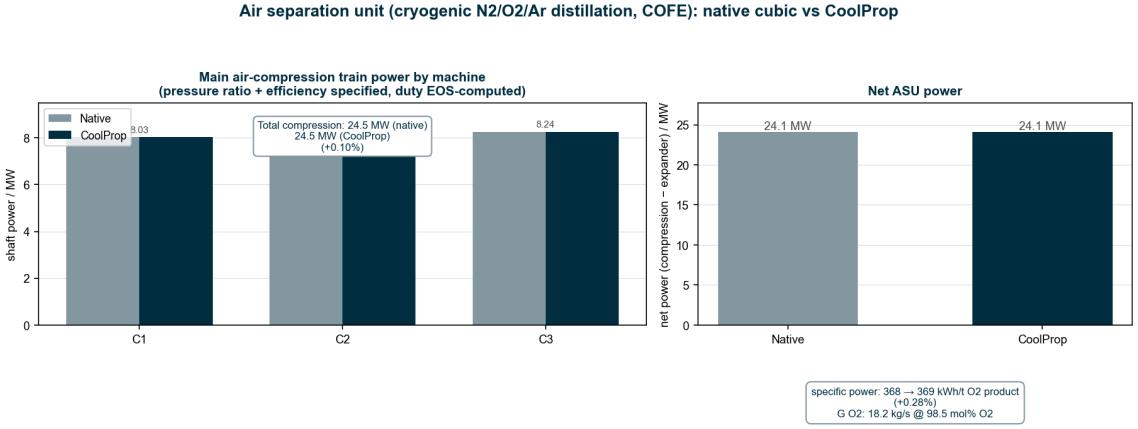


Figure 7.10: Main air-compression train shaft power by machine, net ASU power and specific power per tonne of oxygen product: native cubic (TEA) vs CoolProp.

Table 7.5: ASU: native cubic vs. CoolProp

Metric	Native	CoolProp	Difference
C1 / C2 / C3 compressor power [MW]	8.03 / 8.22 / 8.24	8.03 / 8.23 / 8.26	$< 0.15\%$
Total compression [MW]	24.49	24.52	$+0.10\%$
Expander recovered [MW]	0.393	0.421	$+7\%$ (small abs.)
Net power (compression - expander) [MW]	24.10	24.09	$< 0.02\%$
Specific power [kWh/t O2 product]	367.7	368.8	$+0.28\%$
G O2 flow [kg/s] (98.5 mol% O2)	18.20	18.15	-0.3%
G N2 flow [kg/s] (100 mol% N2)	13.72	13.69	-0.3%
L Ar flow [kg/s] (99.99 mol% Ar)	0.328	0.332	$+1.3\%$

Agreement is within a few tenths of a percent on the compression power, the net power and the major product flows, with the product purities (O2, N2, Ar) identical to specification. The largest relative deltas fall on the small absolute quantities — the recovered expander power and the trace liquid-argon flow — where a small EOS difference in the cryogenic enthalpies and relative volatilities shows up proportionally larger. As with the other mixtures, the differences are the signature of two genuinely different equations of state evaluated correctly, not an interface error: the tight N2/O2/Ar boiling points make this the most demanding separation of the set, and the wrapper still reproduces the native column solution to well within engineering tolerance.

7.6 CO_2 compression, dehydration and liquefaction (CCS)

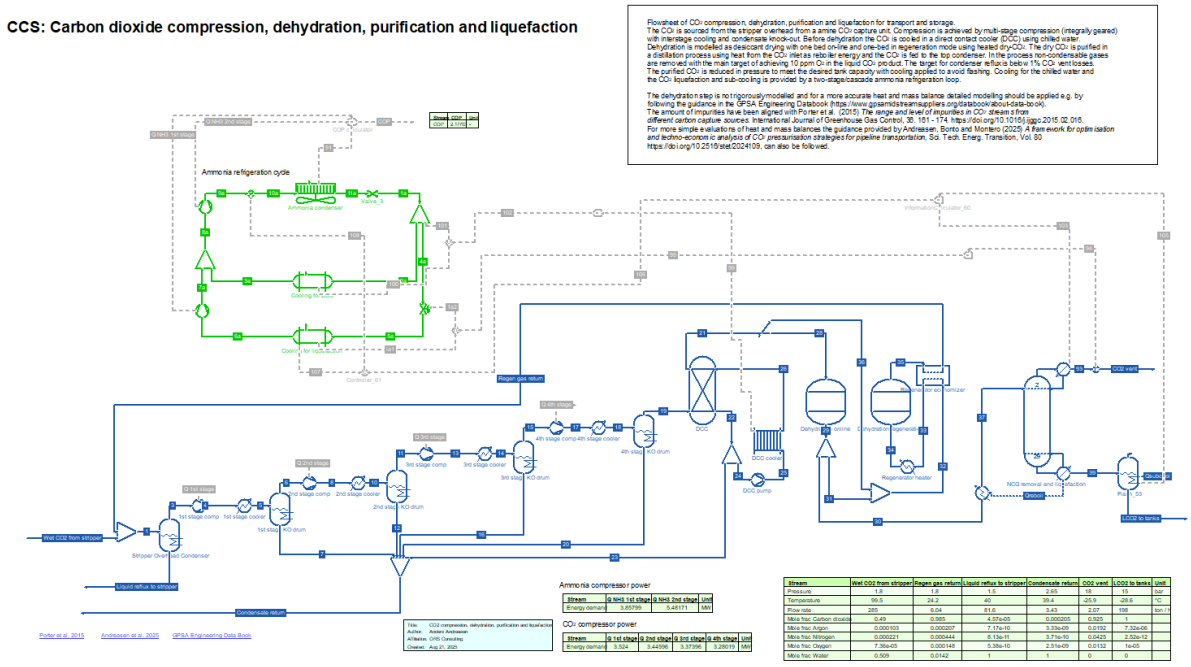


Figure 7.11: COFE flowsheet of the CO_2 compression, dehydration, purification and liquefaction (CCS) process: multi-stage integrally-geared CO_2 compression with interstage cooling and knock-out, a direct-contact cooler, desiccant dehydration, distillation-based non-condensable-gas removal and liquefaction, with a two-stage cascade ammonia refrigeration loop (green). Diagram: `cocosimulator.org/downloads/CO2_compression_liquefaction_Andreasen_2025.png`.

The author's CO_2 compression, dehydration, purification and liquefaction design: multi-stage integrally-geared CO_2 compression with interstage cooling and knock-out, a direct-contact cooler (DCC), desiccant dehydration, distillation-based non-condensable-gas removal and liquefaction, served by a two-stage cascade ammonia refrigeration loop. The feed is a wet, impure CO_2 stream (CO_2/Ar/N_2/O_2/H_2O) and the product is liquid CO_2 at 15 bar delivered to tanks. This is the CCS case of the set, and the only one in which the switched package must span the full range from bone-dry dense-phase CO_2 to near-pure liquid water inside a single compound set.

Unlike the other five flowsheets, this is **not** a whole-flowsheet package swap. COFE binds a property package to a *material type*, and here only the `Carbon dioxide` type — 39 of the 60 material streams — switches from the native cubic to CoolProp. The DCC circulating water loop (7 streams, its own `DCC` type) and the ammonia refrigeration loop (14 streams) stay on the native cubic in **both** cases, the ammonia loop thereby acting as a near-control. The CoolProp case also carries three `StreamConverter` units at the material-type boundary, so the two files are not structurally identical (60 streams / 50 units vs 57 / 47); quantities that straddle that boundary — notably the water condensate — inherit a converter re-flash and are the least clean rows in the table below.

Both files are solved from their saved state (no reset), as in normal COFE use.

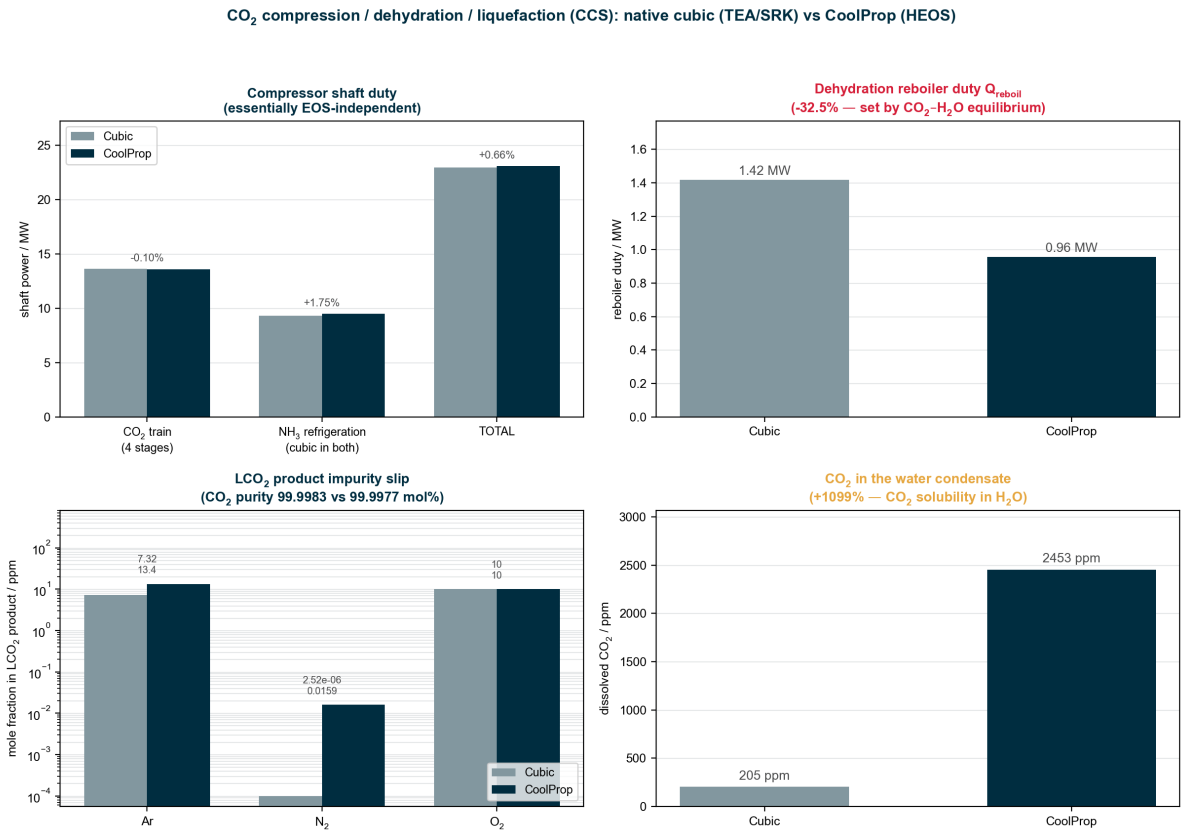


Figure 7.12: CO_2 compression / dehydration / liquefaction solved in COFE with the native cubic (TEA/SRK) and with CoolProp (HEOS): compressor shaft duty (top left), dehydration reboiler duty (top right), LCO_2 product impurity slip (bottom left) and CO_2 dissolved in the water condensate (bottom right).

Table 7.6: CO_2 compression/liquefaction: native cubic vs. CoolProp

Metric	Native	CoolProp	Difference
CO_2 compressor train, 4 stages [MW]	13.624	13.611	-0.10%
NH_3 refrigeration (cubic in both) [MW]	9.340	9.503	$+1.75\%$
Total shaft power [MW]	22.964	23.114	$+0.66\%$
Dehydration reboiler duty [MW]	1.417	0.956	-32.5%
LCO_2 product: CO_2 [mol%]	99.9983	99.9977	-0.0006 pt
LCO_2 product: Ar [ppm]	7.32	13.44	$+84\%$ (small abs.)
LCO_2 product: N_2 [ppm]	~ 0	0.016	trace either way
LCO_2 product flow [kg/s]	54.977	54.951	-0.05%
CO_2 vent: CO_2 [mol%]	92.51	92.36	-0.16%
Condensate: dissolved CO_2 [ppm]	205	2453	:math:` imes 12`
Condensate return: molar flow [mol/s]	52.80	67.12	$+27\%$

The EOS choice separates the results exactly where the underlying phase equilibria differ, and the split is the same one the GERG-2008 study of this flowsheet reports:

- **Compressor shaft duty is essentially EOS-independent.** The CO_2 train agrees to 0.10 % and the product flow to 0.05 %. Compressing dense CO_2-rich gas and delivering a fixed LCO_2 rate is a robust calculation — reassuring for energy and equipment sizing.
- **The dehydration reboiler duty is highly EOS-sensitive**, falling by a third (1.42 → 0.96 MW). The stripper duty is set by the CO_2–H_2O equilibrium, so it is the single most model-dependent number in the flowsheet.
- **The water condensate carries ~12× more dissolved CO:sub:`2`** (205 → 2453 ppm) with a 27 % larger return flow — the direct signature of a different CO_2–H_2O mutual-solubility model. This row straddles the package boundary and should be read as indicative rather than a like-for-like comparison. It is also the row most sensitive to the CoolProp flash revision: the previous bundled build (`c10fc2ac`) gave 1148 ppm and 77.7 mol/s, and needed TP-flash fallbacks in all four KO drums that the current build's liquid-liquid stability probe (CoolProp PR #3473) largely avoids. Every other row moved by less than 0.4 %.
- **Product purity is preserved.** CoolProp slips marginally more Ar (13.4 vs 7.3 ppm) and a trace of N_2, but the LCO_2 product remains at 99.998 mol% CO_2 — comfortably inside any CCS specification.

Read against the published GERG-2008 comparison of this same flowsheet, where the native cubic column reproduces here digit-for-digit, CoolProp lands close to the EOS-CG/Beckmüller "mutant" rather than to stock GERG-2008: reboiler duty 0.96 MW (mutant 0.96, stock GERG 0.72), Ar slip 13.4 ppm (mutant 13, stock GERG 117) and N_2 slip 0.016 ppm (mutant 0.02, stock GERG 94). For a CCS purity specification that difference is decisive, and it is the CO_2–Ar/N_2 binaries — not the pure-fluid EOS — that drive it.

7.7 Summary

Table 7.7: Headline agreement across the six validation flowsheets

Flowsheet	Compound case	Headline metric	Native -> CoolProp	Cold-start
CHP	pure water (*Single-component (pure) compositions*)	net shaft power	agreement $< 0.1\%$	OK
TEALARC	5-compound mixture, exact compound-order match	total refrigeration power	-2.0% (EOS difference, not error)	OK
CO2-from-NG	7-compound, wide boiling range, near-azeotrope	product purity / recovery	$\sim 1.5-1.7$ pt (EOS difference, not error)	needs warm start — *Cold-start initialization on hard mixtures*
Snøhvit	7/8-compound mixture (water-omission caveat above)	net MR train power	$+0.7\%$ (EOS difference, not error)	OK
ASU	3-compound cryogenic (N2/O2/Ar, tight boiling points)	net power / product purity	$< 0.3\%$ (EOS difference, not error)	OK
CO2 compression (CCS)	5-compound, dry CO2 to near-pure water in one set	total shaft power / LCO2 purity	$+0.7\%$ / -0.0006 pt (EOS difference, not error)	warm start (partial swap — see *CO2 compression, dehydration and liquefaction (CCS)*)

All six flowsheets converge cleanly (COFE `SOLVE: OK`) with the compound-order-aligned package (*Compound-order alignment to the Material Object*), and every comparison metric differs from the native reference by an amount consistent with a genuine, expected equation-of-state difference — never by the order-of-magnitude, wrong-phase discrepancies that the pre-fix compound-order bug produced. That distinction is the actual claim of this chapter: not that CoolProp and a native cubic package agree exactly

(they should not — they are different equations of state), but that the property-package interface delivers whichever EOS is selected *correctly* into a converged, real flowsheet solve. One flowsheet (CO2-from-NG) additionally surfaced a genuine, documented cold-start limitation of the HEOS mixture solver with a confirmed, practical workaround (*Cold-start initialization on hard mixtures*) — included here deliberately rather than quietly worked around, since a limitation with a known fix is exactly the kind of finding this validation exercise exists to surface.

Part IV

Appendices

Chapter 8

Appendix: Options and Property Identifiers

8.1 Configuration options

Set via the `Edit()` GUI (*Architecture*) and persisted with the flowsheet (*Design Decisions*).

Option	Values	Notes
Equation of state	PR / SRK / HEOS	*Equation-of-state / backend selection*. One EOS per package instance.
PH/PS flash strategy	C# (default) / Native	*PH/PS flash strategy*. Override for diagnostics: `COOLPROPCO_PHPS=native` / `=csharp` env var.
Compound selection	any subset of the CoolProp fluid catalogue	Realigned to the Material Object's own order by CAS at every calculation call — *Compound-order alignment to the Material Object*.
Binary interaction parameters (kij)	per-pair override	Keyed `"CAS1|CAS2"` in the persisted stream; applied via `AbstractState_set_binary_interaction_double(..., "kij", value)`. Silently ignored if the selected backend does not accept a kij override for that pair (e.g. some HEOS pairs use a different mixing parameter set entirely).

8.2 Diagnostic environment variables

Not part of the persisted flowsheet state — process-level, for troubleshooting and CI.

Variable	Values	Effect
`COOLPROPCO_LOG`	`info` / `debug` / `all`, `off` / `none` / `silent`, unset	Verbosity of `coolpropco.log` (next to the assembly). Unset = errors only (the hot-path default — verbose logging touches disk on every flash).
`COOLPROPCO_PHPS`	`native` / `csharp`	Overrides the configured/persisted PH/PS strategy (*PH/PS flash strategy*) without touching flowsheet state.

8.3 Supported CAPE-OPEN property identifiers

Only identifiers in the CAPE-OPEN 1.1 standard registry (§7.5.5/§7.5.6) may be advertised — a PME (COFE in particular) validates every entry in `GetSinglePhasePropList`/`GetTwoPhasePropList` against that registry and raises "no such property" on anything else. CoolProp-only quantities without a standard CAPE-OPEN identifier (ideal-gas C_v, bulk modulus, isobaric expansion coefficient, isentropic exponent) are therefore **not** advertised, even though the engine can compute them internally.

Table 8.1: Single-phase properties (`CalcSinglePhaseProp`)

Identifier	Source
`enthalpy`, `entropy`	CoolProp `Hmolar`/`Smolar` (or mass basis)
`internalEnergy`, `gibbsEnergy`, `helmholtzEnergy`	CoolProp `Umolar`/`Gmolar`/`Helmholtzmolar`
`heatCapacityCp`, `heatCapacityCv`	CoolProp `Cpmolar`/`Cvmolar`
`volume`, `density`	CoolProp `Dmolar` (reciprocal for volume)
`compressibilityFactor`	CoolProp `Z`
`fugacityCoefficient`, `logFugacityCoefficient`	CoolProp `fugacity_coefficient` per compound
`viscosity`, `thermalConductivity`	CoolProp native, else *Transport-Property Fallbacks* fallback correlations
`speedOfSound`	CoolProp `speed_of_sound`
`molecularWeight`	Mixture-averaged from compound constants
`idealGasHeatCapacity`, `idealGasEnthalpy`, `idealGasEntropy`	CoolProp `Cp0molar`/ideal-gas reference state (needed by COFE for reference-state harmonisation)
`jouleThomsonCoefficient`	Numerical $(\partial T/\partial P)_H$ via CoolProp's partial derivative API
`compressibility`	**Isothermal compressibility** [1/Pa] — note the CAPE-OPEN standard identifier is `compressibility`, not "isothermalCompressibility"

Table 8.2: Two-phase properties (`CalcTwoPhaseProp`)

Identifier	Source
`kvalue`, `logKvalue`	$K_i = y_i/x_i$ from the flash split
`surfaceTension`	CoolProp's native two-phase surface tension (bubble-point state), else the Brock-Bird fallback (*Surface Tension*)

Table 8.3: Compound constants (`GetCompoundConstant`)

Identifier	Notes
`molecularWeight`	kg/kmol (converted from CoolProp's kg/mol)
`criticalTemperature`, `criticalPressure`, `criticalVolume`	
`criticalCompressibility`	Computed from $P_c V_c/(RT_c)$ when V_c is known
`acentricFactor`, `normalBoilingPoint`	
`casRegistryNumber`, `chemicalFormula`	The CAS number is also the key used for compound-order alignment (*Compound-order alignment to the Material Object*) and kij lookup

Only `idealGasHeatCapacity` is offered as a T-dependent compound property; no P-dependent compound properties are provided (`GetPDependentPropList` returns empty/`VT_EMPTY`).

Chapter 9

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Bibliography

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